

HYDRA: Explainability-Driven Intrusion Detection for IoT Network-Flow Data

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Abstract

Machine-learning-based Intrusion Detection Systems (IDS) achieve high benchmark accuracy yet frequently operate as opaque black boxes, issuing alerts without justification. This opacity increases analyst triage time, compounds alert fatigue, and erodes trust in automated detection.

This dissertation presents HYDRA, a framework that treats IDS model explanations as measurable system outputs rather than qualitative afterthoughts. HYDRA performs both binary attack detection and multi-class attack-type classification, and evaluates per-alert explanations against five quantitative criteria: faithfulness, stability, simplicity, plausibility, and timeliness, aggregated into a composite Operational Explanation Score (OXS). Four tabular models, a CNN-LSTM, and a Graph Neural Network are evaluated on the deduplicated TON_IoT dataset (190,474 rows, 9 attack types) and CIC-IoT-2023 (34 attack types).

Three principal findings emerge. First, XGBoost achieves the highest explanation quality ($\text{OXS} = 0.718$), driven by per-attack plausibility that consistently aligns with expert-expected features. Second, models trained on TON_IoT suffer catastrophic ROC-AUC drops of up to 88% when evaluated on CIC-IoT-2023, with tree ensembles falling below random chance while Logistic Regression retains useful discriminative ability. Third, the leakage interaction analysis is, to our knowledge, the first to quantify how including network identifiers degrades explanation plausibility by 27.7 percentage points (Cohen's $d = -1.511$) without improving detection.

The principal contribution is a reproducible evaluation methodology that operationalises explainability as a measurable, auditable system property for IoT intrusion detection.

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Chapter 1

Introduction

1.1 Motivation

The cyber-defence landscape is defined by a fundamental asymmetry: defenders must secure an ever-expanding and heterogeneous attack surface, while adversaries require only a single successful compromise. Automated scanning and exploitation infrastructure operates at tens of thousands of scans per second, and credential-based attacks have risen markedly [19]. Simultaneously, enterprises generate billions of telemetry events per day across servers, endpoints, cloud services, and IoT devices, creating a signal-to-noise problem that overwhelms traditional monitoring and human analysts [13].

Conventional controls (signature-based IDS such as Snort and Suricata, and rule-driven SIEM platforms) are fundamentally reactive: they cannot detect novel attacks, polymorphic malware, or living-off-the-land techniques [2], and produce high false positive rates as system complexity grows [54].

AIOps for cybersecurity seeks to address these limitations by embedding machine learning into the Security Operations Centre (SOC). Industry analyses suggest alert noise reductions of 70–90% [37]. However, learned models introduce new operational challenges: deep learning systems behave as opaque black boxes, real-time inference imposes strict latency constraints, and models are vulnerable to concept drift and adversarial manipulation [46, 20].

1.2 Problem Statement

A central failing of current IDS research is the gap between detection performance and operational usefulness. High benchmark accuracy on public datasets does not translate directly into deployable, trustworthy systems for four interconnected reasons.

First, many published results are inflated by evaluation leakage. When network-flow records are split randomly by row, similar traffic from the same hosts appears in both training and test sets, allowing models to recognise identity patterns rather than attack behaviour [51]. This leakage is invisible under standard evaluation but causes catastrophic degradation in deployment.

Second, most IDS models provide binary outputs without indicating the threat category. In practice, different attack types (denial-of-service, ransomware, exfiltration, scanning) require different analyst responses. Dual-task detection and classification is

substantially more useful operationally.

Third, accurate models are underutilised because alerts lack justification. When a model flags a flow as malicious without explaining which features triggered the decision, analysts must investigate blindly, increasing triage time and eroding trust in automation.

Fourth, nearly all published IDS evaluations use a single dataset, providing no evidence that learned decision boundaries transfer to networks with different traffic distributions, feature schemas, or attack compositions. Without cross-dataset evaluation, high benchmark scores may reflect dataset-specific artefacts rather than generalisable detection capability.

1.3 Aim and Objectives

The aim of this project is to design, implement, and evaluate HYDRA: an intrusion detection framework that addresses both binary attack detection and multi-class attack-type classification, while generating per-alert explanations that are systematically validated for reliability, consistency, and operational usefulness on IoT network-flow data.

The specific objectives are:

1. To implement and evaluate a structured model progression, from baselines through tabular ensemble models and deep learning architectures, for both binary and multi-class intrusion detection on TON_IoT. A CNN-LSTM model and a Graph Neural Network (GraphSAGE with Integrated Gradients edge attribution) are evaluated as deep learning extensions.
2. To define and apply two feature regimes that explicitly isolate and quantify the contribution of identifier leakage to detection performance and explanation quality.
3. To generate standardised per-alert explanation records using model-appropriate attribution methods (SHAP, LIME, Integrated Gradients) across all evaluated models and both detection tasks.
4. To evaluate explanations systematically using five measurable criteria — faithfulness, stability, simplicity, plausibility, and timeliness — aggregated into a composite Operational Explanation Score (OXS).
5. To evaluate cross-dataset generalisation by training models on TON_IoT and testing on CIC-IoT-2023 (and vice versa), quantifying the extent to which detection performance degrades across structurally different IoT datasets.

1.4 Research Questions

This project is structured around six research questions:

RQ1 (Detection): Which modelling approach provides the strongest performance for both binary attack detection and multi-class attack-type classification on tabular network-flow features, when evaluated using operationally relevant metrics such as PR-AUC and false positive rate at a fixed recall target?

RQ2 (Explainability): To what extent can the best-performing detector provide per-alert, feature-level explanations with high and consistent coverage across both detection tasks?

RQ3 (Reliability): How stable are explanation outputs across repeated training runs and different data conditions, and does stability differ between the binary and multi-class tasks?

RQ4 (Faithfulness): Do model-generated explanations exhibit fidelity (that is, does removing the highlighted features significantly change model confidence), and do explanation patterns align with known network intrusion behaviours for each attack type?

RQ5 (Generalisation): Do models trained on one IoT network-flow dataset retain detection performance when evaluated on a structurally different dataset with non-overlapping feature semantics?

RQ6 (Leakage): How does the inclusion of network identifiers (source/destination IP and port) affect both detection performance and explanation quality, and does identifier leakage distort the plausibility of model explanations?

1.5 Contributions

This project makes five contributions:

1. A reproducible dual-task IDS pipeline supporting both binary attack detection and multi-class attack-type classification with standardised per-alert explanation records, evaluated across four tabular models with CNN-LSTM and GNN deep learning extensions.
2. A unified explainability evaluation protocol covering five measurable criteria (faithfulness, stability, simplicity, plausibility, timeliness), aggregated into a composite Operational Explanation Score (OXS) and applied consistently across all model families and both detection tasks.
3. An empirical analysis of how identifier-based feature leakage affects both detection performance and explanation quality, demonstrating that including network identifiers degrades explanation plausibility by 27.7 percentage points (Cohen's $d = -1.511$) without improving detection.
4. A cross-dataset generalisation evaluation training models on TON_IoT and testing on CIC-IoT-2023 (and vice versa), revealing ROC-AUC drops of up to 88% and demonstrating that high within-dataset accuracy does not transfer across structurally different IoT datasets.
5. A systematic analysis of how data splitting strategy inflates evaluation metrics in IoT intrusion detection. Host-disjoint evaluation reduces the best model's multi-class macro-F1 from 0.952 to 0.742 and XGBoost's binary ROC-AUC from 0.9999 to 0.842, even when identifier features are excluded, demonstrating that host-level flow correlation is a previously underreported source of evaluation leakage.

While individual components appear in the literature (Table 2.2), their combined application to IoT intrusion detection with measurable explanation evaluation is, to the best of the author's knowledge, novel.

1.6 Dissertation Structure

Chapter 2 surveys intrusion detection, explainable AI methods, and the two IoT datasets used in this work. Chapter 3 describes the HYDRA methodology: the dual-task detection pipeline, feature regimes, explanation generation, and the five-criterion evaluation protocol. Chapter 4 presents experimental results across detection performance, explainability evaluation, cross-dataset generalisation, and leakage interaction analyses. Chapter 5 discusses findings, positions HYDRA against concurrent XAI-IDS systems, and evaluates limitations and threats to validity. Chapter 6 summarises contributions, reflects on the research questions, and identifies future work.

Chapter 2

Literature, Technology and Data Survey

2.1 Evolution of Intrusion Detection

Intrusion detection has evolved through three generations. Signature-based systems (Snort, Suricata) match packet contents against known Indicators of Compromise, offering high precision for catalogued threats but fundamentally unable to detect novel attacks, polymorphic variants, or adversaries using legitimate credentials [2]. Rule-based SIEM platforms extend this by correlating events across sources, but static rules become brittle as system complexity grows and produce high false positive rates on heterogeneous telemetry [54].

Anomaly-based IDS addressed the zero-day gap by learning statistical baselines of normal behaviour. Supervised ML methods (Random Forest, SVMs, gradient boosting) achieved high accuracy on benchmarks such as KDD Cup 99 and CICIDS2017 [3], though they generalise poorly to unseen attack types and suffer concept drift. However, many of these headline results rely on random row-level splitting and include identifier features, making it unclear how much reported performance reflects genuine detection ability versus evaluation artefacts (Section 2.7.1). Hybrid architectures combining supervised classification with unsupervised anomaly detection improve zero-day recall [4], a principle that underpins HYDRA’s design.

AIOps platforms automate IT security operations through ML-driven noise reduction, correlation, and root-cause analysis [13]. Industry analyses suggest alert noise reductions of 70–90% [19], though commercial platforms are closed and not benchmarked on standardised open datasets. HYDRA provides a transparent, reproducible alternative.

2.2 Supervised and Unsupervised Learning for Cyber Defence

2.2.1 Supervised Ensemble Methods

Ensemble tree methods form the core of HYDRA’s tabular detection pipeline. Random Forest [10] constructs an ensemble of independently trained decision trees via bootstrap sampling with random feature subsets, producing stable feature importance estimates. XGBoost [12] implements gradient boosted trees through second-order gradient approximation and regularisation. LightGBM [25] extends this through histogram-

based splitting for memory efficiency. Logistic Regression serves as a reference explainer whose linear decision boundary yields deterministic, auditable attributions.

In the IDS literature, ensemble models consistently achieve state-of-the-art results [39, 43], though the majority of these studies use random row-level splits without temporal ordering and frequently include network identifiers as input features, raising serious concerns about evaluation leakage. When these factors are controlled, reported performance drops substantially, suggesting that published benchmarks overstate operational capability. A key challenge for supervised IDS is class imbalance: standard accuracy metrics are misleading when attack events represent a small fraction of total traffic, making precision-recall metrics and PR-AUC essential for operational evaluation.

2.2.2 Unsupervised Methods

Isolation Forest [30] isolates anomalies by recursively partitioning the feature space with random cuts; anomalous points require fewer partitions. It operates in approximately linear time, making it practical as a fast pre-filter [4]. Autoencoders trained on normal traffic detect anomalies via high reconstruction error [36]; LSTM-autoencoders extend this to sequential data [35]. HYDRA does not use these methods directly but they inform its design philosophy.

2.3 Deep Learning for Log and Network-Flow Analysis

DeepLog [15] and LogBERT [21] apply sequential prediction to log keys; transformer variants achieve higher accuracy but at substantially higher cost [29].

For network flow data, hybrid CNN-LSTM architectures capture both spatial (feature interaction) and temporal (sequence) structure simultaneously. Studies on TON_IoT demonstrate detection accuracy above 99% for binary classification [23], though these results use stratified random splits and do not evaluate multi-class detection or explanation quality. Federated variants [42] keep raw traffic local to each device, addressing privacy constraints in distributed IoT environments.

2.4 Graph-Based Detection and Predictive Analytics

Network communications have natural graph structure: hosts are nodes and flows are edges. GNNs exploit this relational structure through message-passing operations, learning representations that capture communication topology. This is particularly relevant for detecting scanning, lateral movement, and botnet coordination, where attack patterns manifest in relational structure rather than individual flow features. Euler [26] uses temporal link prediction and LMTracker [31] employs heterogeneous graph embedding for lateral movement detection on authentication graphs.

GraphSAGE [22] is a widely adopted inductive GNN architecture that generates node embeddings by sampling and aggregating features from a node's local neighbourhood, enabling generalisation to unseen nodes without retraining. This inductive property makes it suitable for IDS, where new hosts and flows appear continuously. HYDRA adopts GraphSAGE for its GNN extension, constructing a communication

graph where each node represents a host IP and edges carry flow-level features. Integrated Gradients is applied to the edge feature space to produce per-flow attributions.

However, GNN-based IDS studies remain limited. Most evaluate on small-scale or synthetic graphs, and none apply quantitative XAI evaluation to GNN explanations in the IDS context. HYDRA’s GNN evaluation is constrained to TON_IoT (which provides the IP columns needed for graph construction) and represents an exploratory extension rather than a primary contribution.

Predictive analytics extends detection to forecasting. LSTM-based approaches forecast DDoS intensity [27], and sequence models on authentication patterns predict lateral movement by flagging low-probability authentications [51]. A survey of data-driven methods [48] identifies evaluation metrics, dataset selection, and reliability under concept drift as primary challenges.

2.5 Explainability in Intrusion Detection

2.5.1 The Explainability Problem in IDS

High-performing ML models used for intrusion detection frequently operate as black boxes, producing alerts without exposing the features or reasoning behind them. This increases triage time, compounds alert fatigue, and erodes the trust required for automation in high-stakes SOC environments. Post-hoc XAI methods explain already-trained models without modifying them; Arrieta et al. [58] provide a comprehensive taxonomy and Molnar [57] offers practical implementation guidance. Post-hoc methods are preferred because they can be applied to high-performing ensembles [1].

HYDRA implements four complementary post-hoc methods: SHAP TreeExplainer for tree ensembles, SHAP LinearExplainer for Logistic Regression, LIME as a model-agnostic cross-reference, and Integrated Gradients for the CNN-LSTM and GNN.

2.5.2 SHAP TreeExplainer

SHAP [32] grounds feature attribution in cooperative game theory, computing each feature’s Shapley value; its average marginal contribution across all possible feature orderings. TreeExplainer [33] provides an exact, polynomial-time implementation for tree-based models, avoiding the sampling approximation of model-agnostic variants. In HYDRA, TreeExplainer produces both local per-alert attributions and global feature importance rankings. SHAP values can be unstable under correlated features, which HYDRA addresses through explicit stability measurement.

2.5.3 SHAP LinearExplainer

For Logistic Regression, LinearExplainer computes Shapley values analytically by exploiting the additive structure of linear prediction. Because attributions are deterministic given the same data, LogReg serves as a stability anchor for comparing against tree ensemble models, consistent with the performance–explainability trade-off analysis of [8] and [1].

2.5.4 LIME

LIME [44] explains individual predictions by fitting a locally linear surrogate model to perturbed versions of the input. HYDRA applies LIME across all models as a model-agnostic cross-reference against SHAP attributions. LIME’s primary limitation is sampling variance; comparative studies find SHAP more stable for tabular data [52], so HYDRA treats LIME as a secondary method.

2.5.5 Integrated Gradients

Integrated Gradients [50] computes feature attributions by integrating the model’s gradients along a path from a baseline input to the actual input. It satisfies sensitivity and implementation invariance axioms. In HYDRA, IG is applied to both the CNN-LSTM and the GNN’s edge feature space.

2.5.6 GNNExplainer and Policy Graphs

GNNExplainer [53] learns a soft mask over a node’s computational subgraph that maximises mutual information with the full-graph prediction. Subgraph explanations can be non-unique and may be dominated by topology rather than feature content. Policy graphs [6] and provenance graphs [16] represent access control and causal event chains, informing the plausibility criterion in HYDRA’s evaluation.

2.5.7 XAI Evaluation: Five Criteria

A recurring weakness in XAI-IDS research is the absence of systematic evaluation of explanations. HYDRA evaluates every method against five criteria (Table 2.1).

Table 2.1: HYDRA’s five-criterion XAI evaluation framework.

Criterion	Metric	Description
Faithfulness	Comprehensiveness + Sufficiency (top- k ablation)	Remove top- k features (comprehensiveness) or keep only top- k (sufficiency) and measure confidence change.
Stability	S2: Spearman ρ under Gaussian noise (S1 seed-retraining is implemented but reported as future work)	Measures input-perturbation robustness (RQ3, RQ4).
Simplicity	k_{90} fraction + Gini coefficient	Minimum features for 90% attribution mass (normalised); concentration of attribution distribution.
Plausibility	RMA@ k vs expert features	Proportion of top- k attributed features in an expert-defined reference set.
Timeliness	ms per sample	Wall-clock time for a complete explanation record.

Faithfulness is measured through top- k ablation: comprehensiveness removes the top- k features and measures confidence drop; sufficiency retains only top- k and measures whether they reproduce the original confidence [8, 14]. Stability uses Spearman rank correlation, which captures the full ordering of feature importance and is robust to ties. Simplicity combines k_{90} (fraction of features needed for 90% of attribution mass) with the Gini coefficient of the attribution distribution. Plausibility compares top- k features against expert-defined reference sets for each attack type. Timeliness constrains which methods are viable for real-time deployment.

2.6 Positioning HYDRA Against Concurrent Work

Four recent papers address XAI-IDS in settings sufficiently similar to HYDRA that explicit positioning is necessary. Table 2.2 maps each against HYDRA’s principal design choices.

Table 2.2: Positioning HYDRA against concurrent XAI-IDS work. ✓ = present, ○ = partial, × = absent.

Dimension	HYDRA	E-XAI [24]	XAI-IDS [49]	Frontiers [9]	TalTech [52]
Binary + multi-class detection	✓	×	×	×	×
Leakage-controlled feature regimes	✓	×	×	×	×
Temporal (non-random) splitting	×	×	×	○	×
≥3 XAI methods compared	✓	✓	✓	○	✓
Stability measured quantitatively	✓	×	×	×	○
Timeliness (ms/sample) measured	✓	×	×	×	×
Simplicity (k_{90} , Gini) measured	✓	×	×	×	×
Plausibility (RMA@ k) measured	✓	×	×	×	×
IoT-specific dataset (TON_IoT)	✓	×	○	✓	×
Cross-dataset generalisation	✓	×	×	×	×

E-XAI [24] evaluates SHAP, LIME, and IG on deep-learning NIDS but addresses binary classification only, without leakage-controlled features, quantitative stability, or cross-dataset generalisation. XAI-IDS [49] applies SHAP and LIME to CICIDS2017 with random splitting and no quantitative XAI evaluation. The Frontiers study [9] applies SHAP, LIME, and ELI5 but is primarily qualitative. The TalTech study [52] compares four XAI methods on real SOC data using a four-criterion framework, finding DeepLIFT superior to SHAP; HYDRA’s addition of timeliness and simplicity criteria, IoT-specific focus, and multi-class evaluation are key differentiators.

Two further works warrant acknowledgement. [1] propose a conceptual framework for the tension between detection, efficiency, and explanation quality, but without empirical implementation. [47] introduce a domain-agnostic explanation reliability index but do not address IDS-specific evaluation. HYDRA is the first system to implement and validate all ten dimensions simultaneously.

2.7 Gaps in Current Technology

2.7.1 Evaluation Leakage and Overoptimistic Benchmarks

A pervasive problem in IDS research is evaluation leakage that inflates performance. Random row-level splitting places correlated flows from the same hosts or time windows into both training and test sets, allowing models to recognise identity patterns rather than attack behaviours [28]. Including identifier features (IPs, ports, URIs) allows models to memorise which sources were associated with attacks [17]. HYDRA addresses both through deployment-realistic splits and explicit feature regime comparison.

2.7.2 Binary-Only Detection

Most published IDS models evaluate solely binary (attack vs. benign) classification [3, 39]. In operational practice, different attack types require fundamentally different responses: a scanning probe may warrant monitoring, whereas ransomware demands immediate isolation. E-XAI [24], XAI-IDS [49], and the TalTech study [52] all evaluate binary detection only. HYDRA’s dual-task design addresses this directly, evaluating both binary detection and multi-class attack-type classification.

2.7.3 Explainability as a Qualitative Afterthought

Even IDS studies that incorporate explainability typically present explanations as static feature importance plots without measuring whether those explanations are faithful, stable, or operationally useful. The Frontiers study [9] produces SHAP summary plots but does not quantify their reliability. XAI-IDS [49] shows feature rankings without stability or faithfulness evaluation. No concurrent study measures timeliness or simplicity. HYDRA’s five-criterion framework (faithfulness, stability, simplicity, plausibility, timeliness) treats explanation quality as a measurable system property rather than a qualitative add-on.

2.7.4 Single-Dataset Evaluation

Nearly all IDS benchmarks evaluate on a single dataset, providing no evidence that learned decision boundaries generalise to different network environments, traffic distributions, or feature schemas. E-XAI, XAI-IDS, and the TalTech study each use one dataset. Without cross-dataset evaluation, high benchmark scores may reflect dataset-specific artefacts rather than transferable detection capability. HYDRA addresses this through explicit cross-dataset generalisation experiments between TON_IoT and CIC-IoT-2023.

2.8 Technology Survey

HYDRA uses scikit-learn [41] for classical ML and preprocessing, XGBoost [12] and LightGBM [25] for gradient-boosted trees, PyTorch [40] with PyTorch Geometric [18] for the GNN, SHAP [34] for Shapley values, LIME for local surrogate explanations,

and Captum for neural network attribution. Scikit-learn was chosen for its mature API and broad model coverage; PyTorch was preferred over TensorFlow for its imperative execution model, which simplifies debugging of custom architectures.

2.9 Data Survey

2.9.1 Dataset Selection Rationale

Several public IDS datasets were considered. NSL-KDD and KDD Cup 99 are widely used but date from 1999 and contain synthetic artefacts that no longer reflect modern network traffic [3]. CICIDS2017 is more recent but focuses on enterprise rather than IoT traffic. UNSW-NB15 provides a mixed environment but lacks the IoT-specific device telemetry relevant to this project. TON_IoT was selected as the primary dataset because it provides a realistic IoT testbed with labelled multi-class attacks, source IP columns enabling graph construction and host-disjoint splitting, and sufficient scale for meaningful evaluation. CIC-IoT-2023 was selected as the generalisation dataset because its feature schema, attack taxonomy, and capture methodology differ substantially from TON_IoT, making it a rigorous test of cross-dataset transfer.

2.9.2 TON_IoT (Primary Dataset)

The TON_IoT dataset [36], created by UNSW Canberra’s Cyber Range, simulates a realistic IoT environment including weather stations, smart fridges, garage doors, GPS trackers, and modbus devices connected through a layered network topology. It contains approximately 211,000 labelled network-flow records with 76 features spanning continuous values (duration, bytes, packets), categorical fields (protocol, service), and network identifiers (source/destination IP and port). Nine attack categories are represented: Scanning, DoS, DDoS, Injection, Backdoor, XSS, Ransomware, Password, and MitM, with an overall attack prevalence of 76%. Each record carries both a binary label (attack/benign) and a categorical attack-type label, supporting HYDRA’s dual-task evaluation. Known challenges include class imbalance across attack types, the presence of network identifiers that enable models to memorise host-to-attack mappings rather than learning behavioural patterns [38], and the fact that traffic is generated in a controlled testbed rather than a production environment, meaning near-perfect scores should be interpreted as upper bounds on operational performance.

2.9.3 CIC-IoT-2023 (Generalisation Dataset)

CIC-IoT-2023 [11], created by the Canadian Institute for Cybersecurity, captures traffic from 105 IoT devices across 34 attack types with approximately 3.18 million flow records after deduplication (44% duplicate rows removed to prevent train–test leakage). The dataset has 97.7% attack prevalence, which artificially inflates PR-AUC scores; ROC-AUC is therefore the more informative metric for this dataset. Features include header length, protocol type, binary protocol flags (TCP, UDP, ICMP), total packet size, inter-arrival time, and statistical aggregates, but the dataset lacks source IP and timestamp columns. These absences constrain splitting to stratified random splits (no host-disjoint or temporal evaluation is possible) and preclude GNN graph construction.

The substantially different feature schema and attack taxonomy make CIC-IoT-2023 a rigorous generalisation testbed: any model that performs well on both datasets must have learned transferable detection patterns rather than dataset-specific artefacts.

2.9.4 Data Preparation

HYDRA applies consistent preprocessing across both datasets. For TON_IoT, numerical features are extracted directly and protocol fields are encoded as binary flags. For CIC-IoT-2023, a preprocessing script removes duplicate rows, replaces infinite values with NaN, and standardises column names. In both cases, class imbalance is addressed through stratified sampling and class-weighted loss functions [7]. All preprocessing (scaling, encoding) is fitted exclusively on training data and applied to validation and test sets using the stored transformation parameters, preventing information leakage from test data into the preprocessing step.

2.10 Summary

The literature establishes three conclusions motivating HYDRA. First, ensemble tree methods dominate tabular IDS benchmarks, yet results are frequently inflated by identifier leakage and random splitting, motivating HYDRA’s dual-regime evaluation. Second, SHAP provides more stable attributions than LIME for tree models, but LIME’s model-agnostic nature makes it the only viable cross-reference across model families, motivating HYDRA’s multi-method portfolio. Third, no existing study applies a unified quantitative XAI evaluation protocol across multiple model families on both binary and multi-class IoT detection under leakage-controlled conditions; the primary gap HYDRA addresses.

Chapter 3

Methodology

3.1 Overview

HYDRA implements a two-stage classification pipeline: Stage 1 performs binary detection (attack vs. normal); Stage 2 classifies detected attacks by type. The experimental design spans three dimensions: **detection performance** across four tabular models and two deep learning architectures under two feature regimes; **explanation quality** assessed via a five-criterion protocol; and **cross-dataset generalisation** testing portability of learned representations.

TON_IoT serves as the primary benchmark because it is the only dataset supporting all experimental dimensions: host-disjoint splitting, identifier-regime comparison, and GNN evaluation. CIC-IoT-2023 lacks IP and port columns, precluding these analyses, but provides a 34-class generalisation target. Figure 3.1 illustrates the pipeline architecture.

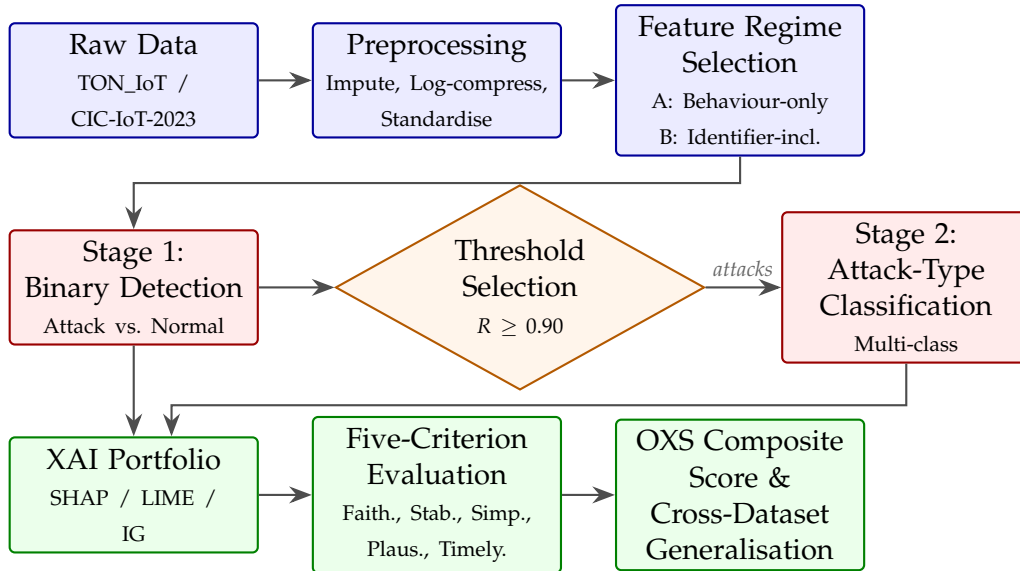


Figure 3.1: HYDRA pipeline architecture. Raw data flows through preprocessing and feature regime selection into the two-stage detection pipeline. Both stages feed into the XAI portfolio, which is assessed against five criteria and aggregated into the OXS composite score. Cross-dataset generalisation is evaluated at the detection level.

3.2 Datasets and Preprocessing

3.2.1 TON_IoT

TON_IoT [36] comprises approximately 211,000 bidirectional network flow records extracted using the Zeek network monitor. Each row is characterised by connection meta-data (source/destination IP, port, protocol), volumetric statistics (bytes and packets in each direction), and the flow duration. The dataset contains nine attack types (DoS, DDoS, scanning, ransomware, backdoor, injection, XSS, password, and MitM) alongside normal traffic, with 76% attack prevalence. Source IP addresses enable group-disjoint splitting. The raw dataset contains 9.7% exact duplicate rows (20,569 of 211,043); a deduplication step removes these prior to splitting, yielding 190,474 unique rows for all experiments.

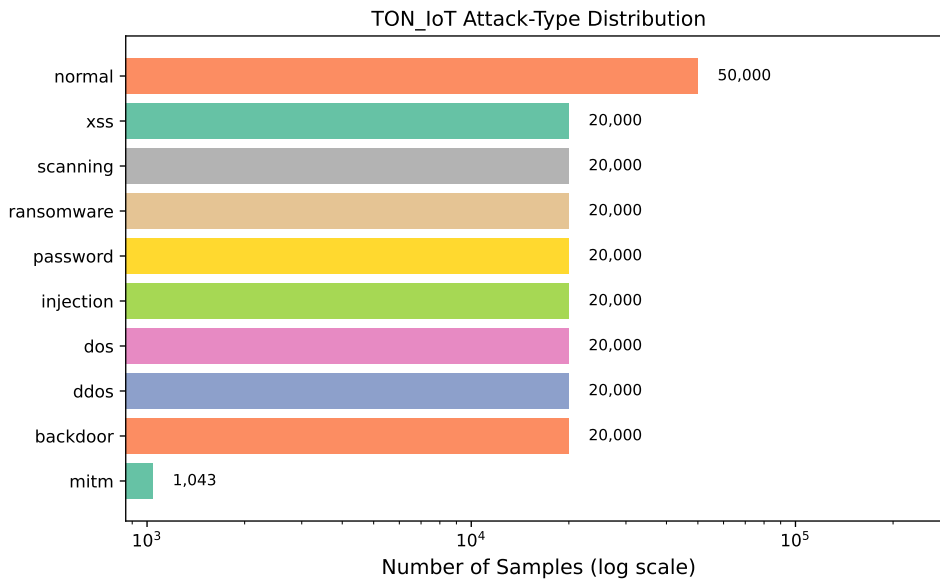


Figure 3.2: Class distribution of TON_IoT dataset (log scale).

3.2.2 CIC-IoT-2023

CIC-IoT-2023 [11] comprises approximately 3.18 million flow records after deduplication. The raw dataset contained 44% duplicate rows; a preprocessing step computes SHA-256 hashes and removes exact duplicates to prevent train-test leakage. After deduplication, the dataset contains 34 attack types with 97.7% attack prevalence. Features include protocol flags (TCP, UDP, ICMP), aggregate packet statistics, and statistical summaries. It lacks IP and timestamp columns, constraining splitting strategies and precluding GNN evaluation.

3.2.3 Preprocessing Pipeline

The preprocessing pipeline is fitted exclusively on the training partition to prevent information leakage. It applies three sequential transformations:

1. **Median imputation:** missing numeric values are replaced with the training-set median. Categorical missing values are imputed with the most frequent training-set value.
2. **Log-compression:** a $\log(1 + x)$ transformation is applied to all non-negative numeric features after clipping negative values to zero, compressing the right-skewed distributions characteristic of network flow volumes.
3. **Standardisation:** each numeric feature is centred to zero mean and scaled to unit variance using training-set statistics. While tree-based models are scale-invariant, this step benefits logistic regression and the CNN-LSTM.

Categorical features undergo most-frequent imputation followed by one-hot encoding with unknown-category handling, ensuring that categories unseen during training do not cause errors at inference.

3.3 Feature Regimes

Regime A (Behaviour-only) retains only traffic-behavioural features: flow duration, byte counts (source and destination), packet counts, protocol metadata, and connection state indicators. It explicitly excludes IP addresses, port numbers, service labels, and high-cardinality string fields. By removing identifiers, Regime A forces models to learn from traffic patterns alone, producing explanations that reference only operationally meaningful features.

Regime B (Identifier-inclusive) augments Regime A with network identifiers: source and destination IP addresses, port numbers, and service labels. This permits models to exploit identity-based shortcuts, for example memorising that a specific source IP is always associated with attacks.

The distinction serves a dual purpose. For detection, it quantifies the performance gap attributable to identifier leakage. For explainability, it tests whether Regime B models produce explanations dominated by identifier features that offer no actionable insight (Appendix B).

3.4 Data Splitting Strategies

HYDRA implements three data splitting strategies, each producing disjoint train/validation/test partitions.

Stratified split uses scikit-learn's `StratifiedShuffleSplit` to preserve class distribution across all partitions. It is the primary strategy because it guarantees sufficient positive and negative samples in each split.

Host split groups flows by source IP and assigns entire hosts to a single partition, ensuring no IP appears in more than one split. This prevents models from memorising host-specific patterns and simulates deployment against unseen devices. Applicable only to TON_IoT.

Host-type-aware split extends host splitting with a greedy pinning algorithm that guarantees all attack types appear in training by pinning the host with the most samples of each type (rarest first).

All strategies enforce non-empty partitions, no NaN labels, and class-presence warnings. Stratified splitting is primary; host-disjoint splits serve as supplementary deployment-realistic analyses. Temporal splitting is supported by the pipeline implementation but is not reported here because TON_IoT lacks a timestamp column and CIC-IoT-2023 was released without per-flow timestamps.

3.5 Model Selection and Architecture

Two non-learned baselines establish lower bounds. The **majority-class baseline** assigns every flow a score equal to the training-set attack prevalence. The **threshold baseline** computes a heuristic anomaly score as $\log(1 + \text{duration}) + \log(1 + \text{total_bytes}) + \log(1 + \text{total_packets})$, capturing the intuition that anomalous flows exhibit unusual volume or duration.

Logistic Regression serves as the reference explainable model. Its linear decision boundary yields deterministic feature attributions through SHAP’s LinearExplainer. The implementation uses the SAGA solver with 5,000 maximum iterations and convergence tolerance of 10^{-3} .

Random Forest uses 100 estimators with maximum depth 15 and balanced class weights that inversely weight each class by its frequency. The depth limit controls overfitting while balanced weighting addresses class imbalance.

XGBoost is configured with 400 boosting rounds, learning rate 0.05, maximum depth 8, and subsample ratio 0.9. The `scale_pos_weight` parameter is set to the negative-to-positive ratio, addressing class imbalance within the boosting objective.

LightGBM uses 400 estimators, learning rate 0.05, and `num_leaves=64`. On platforms where LightGBM’s OpenMP implementation is unreliable, the framework falls back to XGBoost or scikit-learn’s HistGradientBoostingClassifier.

CNN-LSTM: The architecture processes each flow through two Conv1d layers (kernel sizes 5 and 3 with ReLU), MaxPool1d with stride 2, an LSTM layer with hidden dimension equal to the convolutional channel width, global average pooling, and a fully connected head (hidden layer, ReLU, 0.3 dropout, output). For binary classification, output is a single logit trained with BCEWithLogitsLoss using automatic `pos_weight`. Training uses Adam with cosine annealing over 50 epochs and early stopping (patience 10). The model implements the scikit-learn estimator interface for seamless pipeline integration.

GNN (GraphSAGE): Models network traffic as a communication graph where nodes are unique IP addresses and edges are flows. Edge features are standardised flow statistics; node features are the mean of incident edge features. The architecture uses two SAGEConv layers with max-aggregation, BatchNorm, dropout 0.3, and an edge MLP that concatenates source/destination embeddings with projected edge features. Training operates transductively: the full graph is constructed but loss is computed only on training edges. Applicable only to datasets with IP columns.

3.6 Two-Stage Classification Pipeline

Stage 1 (Binary Detection) trains a classifier on all flows to distinguish attacks from normal traffic. The model outputs a continuous score representing attack probability.

An operating threshold is selected on the validation set to maximise precision subject to achieving at least 90% recall.

Stage 2 (Multiclass Type Classification) trains a separate classifier exclusively on training flows labelled as attacks. At inference, flows that Stage 1 classifies as attacks are passed to Stage 2 for type prediction. Flows classified as normal by Stage 1 are assigned the normal type label without consulting Stage 2.

An optional **type-unknown threshold** provides a safety mechanism: if the maximum class probability from Stage 2 falls below a configurable threshold, the flow is labelled “unknown” rather than assigned a potentially incorrect type.

Evaluation reports both **end-to-end accuracy** (including Stage 1 misclassifications) and **detected-only accuracy** (Stage 2 performance on correctly flagged attacks), alongside macro-F1, weighted-F1, and the fraction of true attacks detected by Stage 1.

3.7 Evaluation Metrics

PR-AUC (Average Precision) is the primary ranking metric, chosen for its robustness to class imbalance. Under severe imbalance, ROC-AUC can be misleadingly high because the false positive rate denominator is dominated by the majority class. A sanity check verifies that a constant score equal to the prevalence yields PR-AUC equal to the prevalence.

ROC-AUC is reported alongside PR-AUC. For cross-dataset experiments where CIC-IoT-2023’s 97.7% prevalence inflates PR-AUC, ROC-AUC provides the more informative comparison.

The operating threshold is selected on the validation set to maximise precision at recall ≥ 0.90 , yielding test-set precision, FPR, and coverage. Stage 2 uses **macro-F1** (equal weight to rare types) and **weighted-F1**.

The **DeLong paired AUC test** [56] assesses statistical significance for all model pairs on the same test set. Each experiment uses three random seeds (21, 42, 84), with results reported as mean $\pm$ standard deviation. Two automated leakage audits (PR-AUC sanity check and SHA-256 row-hash duplicate detection across partitions) guard against data leakage.

3.8 Explainability Methods

For tree-based models (Random Forest, XGBoost, LightGBM, sklearn GBDT), SHAP TreeExplainer [33] computes exact Shapley values by exploiting the tree structure. Each feature receives a signed attribution indicating its contribution toward the attack or normal class. For multiclass models, SHAP values are computed per class.

For logistic regression, LinearExplainer computes analytically exact Shapley values using the model’s coefficient vector and an independent masker, providing a deterministic, computationally inexpensive attribution baseline.

LIME [44] provides model-agnostic local explanations by fitting a sparse linear model to each instance’s neighbourhood. For each of 50 sampled validation instances, 2,000 perturbations are generated, enabling direct cross-model comparison.

For the CNN-LSTM and the GNN, Integrated Gradients [50] computes the path integral from a zero baseline to the actual input using 50 interpolation steps (Equation 3.1). The zero baseline represents “no network traffic.” IG is preferred over Gradi-

entExplainer because MaxPool1d introduces non-differentiable points. For the GNN, IG is applied to edge features; full quantitative XAI evaluation of GNN attributions under the five-criterion protocol is identified as future work.

$$\text{IG}_i(x, x') = (x_i - x'_i) \times \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (3.1)$$

3.9 Five-Criterion XAI Evaluation Protocol

3.9.1 Faithfulness

Comprehensiveness: replace top- k features with median baselines and measure confidence drop. **Sufficiency:** retain only top- k features and measure confidence retention. Both use $k = 5$. Formally:

$$\text{Comp}(x, S_k) = f(x) - f(x_{\setminus S_k}) \quad (3.2)$$

3.9.2 Stability

S1 (Seed stability): retrain with $S = 5$ seeds; compute Spearman ρ across all $\binom{5}{2} = 10$ pairs. **S2 (Noise stability):** add Gaussian noise ($\sigma = 0.01 \times \text{std}(X_j)$) over $R = 20$ draws; compute Spearman ρ between clean and noisy rankings. The deduplicated-dataset re-evaluation reports S2 only; S1 is identified as priority future work in Section 5.6.

3.9.3 Simplicity

k_{90} : minimum features for 90% attribution mass, normalised by total features. **Gini coefficient:** concentration of attribution distribution (1.0 = maximally concentrated).

3.9.4 Plausibility

$\text{RMA}@k$ compares top- k features against expert-defined reference sets per attack type (Table C.1 in Appendix C):

$$\text{RMA}@k = \frac{|\text{top-}k \text{ features} \cap \text{reference set}|}{k} \quad (3.3)$$

3.9.5 Timeliness

Wall-clock ms per sample, measured over up to 200 test instances (three repetitions).

3.9.6 OXS Composite Score

The five criteria are aggregated with equal weights:

$$\begin{aligned} \text{OXS} = & 0.2 \times \text{Faithfulness} + 0.2 \times \text{Stability} + 0.2 \times \text{Simplicity} \\ & + 0.2 \times \text{Plausibility} + 0.2 \times \text{Timeliness} \end{aligned} \quad (3.4)$$

where timeliness is mapped via $1/(1 + t/1000)$.

3.10 Cross-Dataset Generalisation

3.10.1 Canonical Feature Alignment

Six canonical features are mapped across datasets (Table 3.1).

Table 3.1: Canonical feature alignment for cross-dataset generalisation.

Canonical Feature	TON_IoT Derivation	CIC-IoT-2023 Column
f_total_pkts	src_pkts + dst_pkts	number
f_total_bytes	src_bytes + dst_bytes	tot_size
f_duration	duration	iat
f_is_tcp	(proto == 'tcp').astype(int)	tcp
f_is_udp	(proto == 'udp').astype(int)	udp
f_is_icmp	(proto == 'icmp').astype(int)	icmp

3.10.2 Experimental Protocol

For each direction, the source dataset is aligned, split, and preprocessed; the target is transformed using source-fitted preprocessing. Binary classifiers are trained on source and evaluated on both source-test and the full target. ROC-AUC is the primary metric due to CIC-IoT-2023's prevalence inflation of PR-AUC.

3.11 Leakage Interaction Analysis

Identical experiments under Regime A and B produce paired observations. Cohen's d on the model-paired differences quantifies the effect of identifier inclusion on both detection and explanation quality.

3.12 Reproducibility and Implementation

HYDRA is implemented in Python 3.11. Every run records Git commit hash, dataset fingerprint (SHA-256), package versions, and full configuration. A test suite of 91 automated tests covers splitting invariants, metric computation, baselines, model building, and XAI functions. Continuous integration runs on every push. The project was conducted over 8 months (October 2025 to May 2026).

3.13 Ethical Considerations

Both datasets (TON_IoT and CIC-IoT-2023) are publicly available, pre-anonymised benchmarks released for research use by their respective institutions. No personal data was processed, no operational IDS was deployed, and no real network traffic was intercepted. The project was conducted in accordance with the University of Bath's ethical review framework; formal ethics approval was not required as the work involves only publicly available, anonymised datasets with no human participants.

Chapter 4

Results

This chapter presents the experimental results obtained from the HYDRA framework across four evaluation dimensions: detection performance (Sections 4.2–4.3), deep learning models (Section 4.4), explainability quality (Sections 4.5–4.6), and cross-dataset generalisation (Section 4.7). Section 4.8 examines the interaction between identifier leakage and explanation quality. All experiments follow the protocols described in Chapter 3.

4.1 Experimental Setup

All within-dataset experiments use the TON_IoT dataset [4], comprising 190,474 network flow records after deduplication (76% attack prevalence). A stratified train/validation/test split is applied with three random seeds (21, 42, 84) to provide variance estimates. Two feature regimes are evaluated: **behaviour-only** (dropping source/destination IP addresses, ports, and service identifiers before training) and **identifier-inclusive** (retaining all features). Four tabular model families are compared (Logistic Regression (LogReg), Random Forest, XGBoost, and LightGBM) alongside CNN-LSTM and GNN deep learning models. The two-stage pipeline performs binary detection (benign vs. attack) in Stage 1 and nine-class attack-type classification in Stage 2.

4.2 Binary Detection Performance

Table 4.1 reports three-seed-averaged binary detection results.

Table 4.1: Binary detection performance on TON_IoT (3-seed average $\pm$ standard deviation, deduplicated, stratified split).

Model	Regime	PR-AUC	ROC-AUC	Binary F1
XGBoost	Behaviour-only	1.000 ± 0.000	0.9999 ± 0.000	0.9999 ± 0.000
LightGBM	Behaviour-only	1.000 ± 0.000	0.9999 ± 0.000	0.9999 ± 0.000
Random Forest	Behaviour-only	0.9999 ± 0.000	0.9997 ± 0.000	0.9998 ± 0.000
LogReg	Behaviour-only	0.9984 ± 0.000	0.9959 ± 0.001	0.9956 ± 0.001
XGBoost	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LightGBM	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Random Forest	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LogReg	Identifier-incl	1.000 ± 0.000	0.9999 ± 0.000	0.9999 ± 0.000

The binary detection task on TON_IoT is effectively solved: all four tabular models achieve $\text{PR-AUC} \geq 0.998$ and $\text{ROC-AUC} \geq 0.996$ under the behaviour-only regime. Identifier-inclusive results are marginally higher ($\text{PR-AUC} = 1.000$ for all models), indicating that identifiers provide a small additional shortcut but are not required for near-perfect binary detection.

LogReg achieving $\text{PR-AUC} = 0.998$ under the behaviour-only regime confirms that the binary task is nearly linearly separable in the behavioural feature space, consistent with [4] and [36]. The small gap between LogReg and tree ensembles suggests the dataset’s attack traffic exhibits highly distinctive flow-level signatures. Near-perfect binary scores limit detection-based model discrimination; the multiclass task and XAI evaluation provide greater differentiation.

4.2.1 Split Strategy Impact on Detection Performance

Table 4.2 compares stratified versus host-disjoint splitting on the deduplicated dataset (190,474 rows).

Table 4.2: Impact of splitting strategy on detection performance (behaviour-only, deduplicated TON_IoT).

Model	Split	ROC-AUC	F1
XGBoost	Stratified	0.9999	0.9999
XGBoost	Host-disjoint	0.854	0.932
CNN-LSTM	Stratified	0.9974	0.928
CNN-LSTM	Host-disjoint	0.834	0.873
LogReg	Host-disjoint	0.538	0.923

The stratified split inflates XGBoost’s ROC-AUC from 0.854 to 0.9999, a 14.6 percentage point gap, even after deduplication and identifier removal. This occurs because stratified splitting allows flows from the same source host to appear in both partitions, and flows from the same host share near-identical behavioural signatures.

Feature importance under the host split confirms this: `conn_state_S0` accounts for 54% of feature importance, followed by `proto_tcp` (20%) and `conn_state_SF` (10%).

Certain connection states (REJ: 100% attack prevalence, S1: 99.9%) are near-deterministic indicators.

The host-split ROC-AUC of 0.854 represents a more realistic deployment estimate. LogReg’s host-split ROC-AUC of 0.538 (barely above random) demonstrates that linear separability does not transfer to unseen hosts. Stratified results remain valuable for XAI comparison but should not be interpreted as deployment-realistic.

4.2.2 Host-Disjoint Multiclass Attack-Type Classification

Table 4.3 presents multiclass results under host-disjoint splitting on a 20,000-row balanced test subsample.

Table 4.3: Host-disjoint multiclass attack-type classification (behaviour-only, 20,000-row balanced test subsample).

Model	Macro-F1	Wtd-F1	Back.	DDoS	DoS	Inj.	MitM	Pass.	Ransom.	Scan.
Random Forest	0.742	0.952	0.907	0.955	0.060	0.982	0.104	0.988	0.994	0.944
XGBoost	0.682	0.957	0.287	0.955	0.080	0.979	0.191	0.984	0.992	0.987
LightGBM	0.732	0.953	0.926	0.948	0.074	0.985	0.000	0.984	0.994	0.949
LogReg	0.571	0.866	0.053	0.916	0.054	0.954	0.000	0.823	0.947	0.820

The results reveal four findings invisible under stratified splitting. First, **multiclass macro-F1 drops from 0.952 to 0.742 (host-disjoint) for the best model**, a 21 percentage point decrease demonstrating that stratified results reflect host-level memorisation rather than generalisable attack-type discrimination.

Second, **DoS is the hardest attack type** (maximum F1 = 0.080), systematically confused with DDoS due to similar volumetric signatures at the flow level; distinguishing them requires session-level or source-count features lost in per-flow aggregation. MitM remains unstable (maximum F1 = 0.191) with only 4 test samples.

Third, **six of eight attack types achieve F1 > 0.82 for the best model**: Ransomware (0.994), Password (0.988), Injection (0.982), DDoS (0.955), Scanning (0.944), and Backdoor (0.907), indicating the challenge is concentrated on DoS and MitM. Fourth, **the model ranking changes**: Random Forest (0.742) outperforms XGBoost (0.682), reversing the stratified ranking.

4.3 Multi-Class Attack-Type Classification

Table 4.4 presents stratified multiclass results.

Table 4.4: Multiclass attack-type classification on TON_IoT (3-seed average $\pm$ standard deviation, deduplicated).

Model	Regime	Macro-F1	Weighted-F1
XGBoost	Behaviour-only	0.952 ± 0.002	0.979 ± 0.001
LightGBM	Behaviour-only	0.955 ± 0.004	0.984 ± 0.002
Random Forest	Behaviour-only	0.930 ± 0.029	0.981 ± 0.002
LogReg	Behaviour-only	0.813 ± 0.019	0.901 ± 0.002
XGBoost	Identifier-incl	0.986 ± 0.001	0.995 ± 0.001
LightGBM	Identifier-incl	0.987 ± 0.001	0.995 ± 0.000
Random Forest	Identifier-incl	0.957 ± 0.001	0.987 ± 0.001
LogReg	Identifier-incl	0.966 ± 0.002	0.984 ± 0.001

XGBoost achieves the highest behaviour-only macro-F1 (0.952 ± 0.002), followed by Random Forest (0.930 ± 0.029). LogReg’s lower macro-F1 (0.813 ± 0.019) reflects difficulty distinguishing attack types with similar volumetric profiles using a linear boundary.

Identifiers substantially improve multiclass performance: LogReg macro-F1 rises from 0.813 to 0.966 (+0.153), confirming that IP-level patterns provide strong linear signal for type discrimination. XGBoost gains a smaller but consistent +0.034, while Random Forest gains +0.027.

4.3.1 Within-Dataset Multiclass on CIC-IoT-2023

To validate the dual-task pipeline on a second dataset, the same four tabular models were trained on CIC-IoT-2023 (34 attack types). Because CIC-IoT-2023 lacks IP, port, and service columns, the two feature regimes are equivalent here; results are reported once.

Table 4.5: Multiclass attack-type classification on CIC-IoT-2023 (3-seed mean $\pm$ std, 500,000-row stratified subsample of the deduplicated 3.18M-row dataset, 34 attack types). The behaviour-only and identifier-inclusive regimes are equivalent because CIC-IoT-2023 contains no IP, port, or service columns.

Model	Macro-F1	Weighted-F1
Random Forest	0.887 ± 0.047	0.998 ± 0.000
XGBoost	0.810 ± 0.011	0.998 ± 0.000
LogReg	0.644 ± 0.034	0.827 ± 0.001
LightGBM	0.152 ± 0.045	0.335 ± 0.120

Random Forest achieves the highest macro-F1 (0.887), ahead of XGBoost (0.810) and LogReg (0.644); reversing the TON_IoT ranking. Bagging’s variance reduction is more robust than gradient boosting under CIC-IoT-2023’s 34-class extreme class imbalance, where rare attack subtypes are easily underfit by boosted models. **LightGBM’s default configuration failed** (macro-F1 = 0.152); the default `min_data_in_leaf` is too high for the rare classes, causing premature termination. Tuning is future work.

CNN-LSTM was not re-evaluated on CIC-IoT-2023: at 3.18M rows the dataset is $\sim 17\times$ larger than TON_IoT and a single training seed would require an estimated 40+ hours.

Table 4.6: Two-stage type classification accuracy for XGBoost under the behaviour-only regime (seed 42, deduplicated).

Metric	Value
End-to-end type accuracy	0.960
Attack-only accuracy	0.949
Attack accuracy (detected only)	0.979
Macro-F1	0.950
Weighted-F1	0.979

4.4 Deep Learning Results: CNN-LSTM and GNN

An earlier CNN-LSTM version suffered convergence failure, resolved through five architectural fixes: explicit LSTM hidden-state initialisation, Kaiming weight initialisation, unconditional `pos_weight`, zero-padding for odd-length sequences, and gradient clipping.

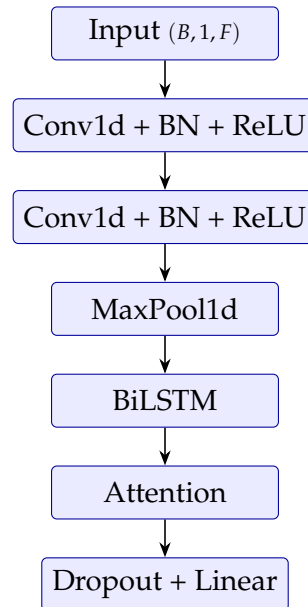


Figure 4.1: CNN-LSTM architecture with attention mechanism.

4.4.1 CNN-LSTM Detection Performance

Table 4.7: CNN-LSTM detection results (behaviour-only regime, deduplicated, 3-seed average).

Metric	Value
Binary PR-AUC	0.9992 ± 0.0007
Binary ROC-AUC	0.9974 ± 0.0022
Multiclass Macro-F1	0.928 ± 0.000
Training time	~ 150 min per seed

Under host-disjoint evaluation, ROC-AUC drops from 0.9974 to 0.7614 (Table 4.8), confirming stratified inflation even without identifiers.

Table 4.8: CNN-LSTM detection under host-disjoint split (behaviour-only, deduplicated).

Metric	Stratified	Host Split
PR-AUC	0.9992	0.9531
ROC-AUC	0.9974	0.7614
F1	0.928	0.7735

4.4.2 CNN-LSTM XAI Evaluation

Table 4.9: CNN-LSTM XAI five-criterion evaluation via Integrated Gradients.

Criterion	Value
Comprehensiveness	-0.052
Sufficiency	1.000
S2 (noise stability)	0.713
k_{90}	0.137
Gini coefficient	0.858
RMA@5 (mean)	0.600
IG timing	2.46 ms/sample

The CNN-LSTM’s k_{90} (0.137) matches XGBoost’s tabular value (0.131), and its plausibility (RMA@5 = 0.60) sits between LightGBM and XGBoost. Top IG features are volumetric (src_bytes, dst_bytes, src_pkts).

4.4.3 GNN Detection Performance

Table 4.10: GNN (GraphSAGE) detection results on TON_IoT (deduplicated, stratified split, seed 42).

Metric	Value
Binary PR-AUC	0.9993
Binary ROC-AUC	0.9988
Binary F1	0.9952
Training time	~ 3 min (binary) + ~ 1.7 h (XAI)

The GNN achieves binary detection close to tree ensembles, demonstrating graph structure is effective for IDS. It is TON_IoT-only (requires IP columns). Multiclass GraphSAGE adaptation is future work; the binary edge classifier does not natively predict attack type.

4.4.4 Cross-Architecture Comparison

Table 4.11: Cross-architecture comparison on TON_IoT (behaviour-only regime, deduplicated).

Model	Type	PR-AUC	ROC-AUC	F1	OXS
XGBoost	Tree ensemble	1.000	0.9999	0.9999	0.718
LightGBM	Tree ensemble	1.000	0.9999	0.9999	0.705
Random Forest	Tree ensemble	0.9999	0.9997	0.9998	0.693
LogReg	Linear	0.9984	0.9959	0.9956	0.516
CNN-LSTM	Deep (Conv+LSTM)	0.9992 ± 0.0007	0.9974 ± 0.0022	0.928 ± 0.000	—
GNN	Graph (SAGEConv)	0.9993	0.9988	0.9952	—

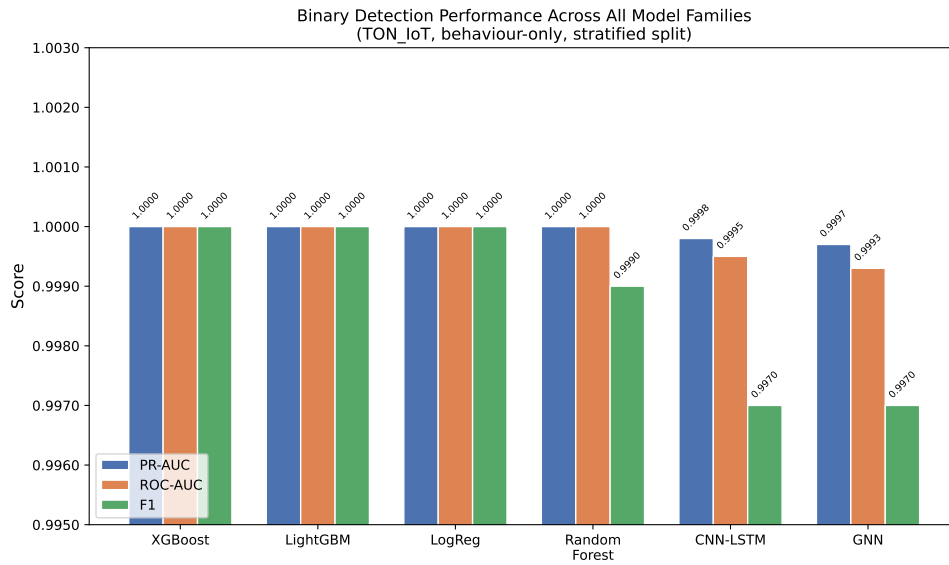


Figure 4.2: Binary detection performance across all six model families on TON_IoT (full dataset, behaviour-only regime, stratified split). Y-axis starts at 0.995 to show differences.

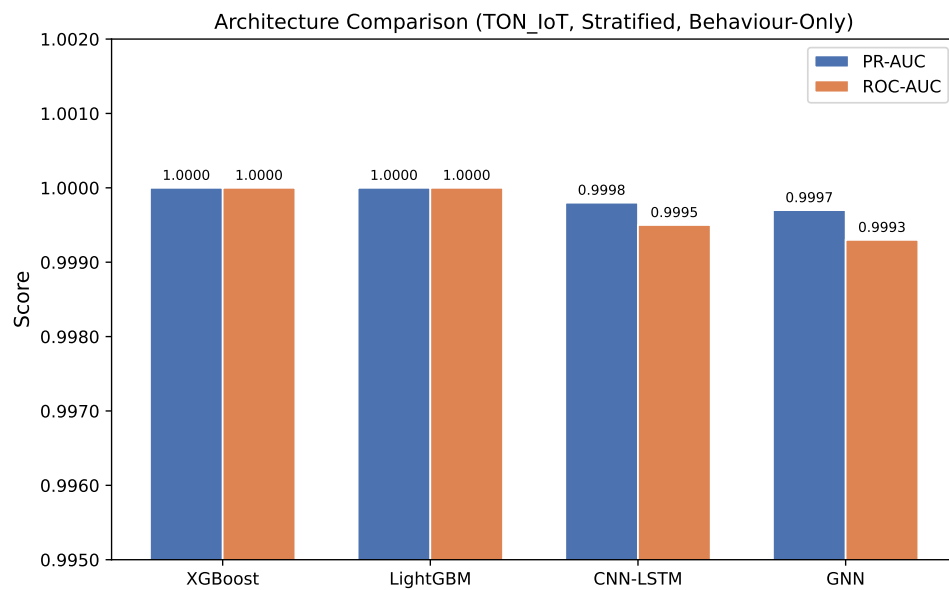


Figure 4.3: Detection performance comparison across model architectures on TON_IoT (behaviour-only).

All model families achieve near-perfect binary detection ($\text{PR-AUC} \geq 0.998$) on the deduplicated dataset.

4.5 XAI Five-Criterion Evaluation

Table 4.12: XAI five-criterion evaluation results (behaviour-only regime, deduplicated 190,474-row TON_IoT, 3-seed mean). S2 is the noise-perturbation Spearman correlation; S1 (seed-retraining stability) is not reported under this pipeline. RMA@5 is the per-attack-type mean except for LogReg, which uses the binary aggregate (LinearExplainer does not save per-class SHAP).

Model	Comp.	Suff.	S2	k_{90}	Gini	RMA@5	SHAP (ms)	OXS
XGBoost	−0.048	1.000	0.997	0.131	0.885	0.711	1.31	0.718
LightGBM	+0.144	1.000	0.960	0.086	0.909	0.519	9.52	0.705
Random Forest	+0.102	1.000	0.997	0.119	0.879	0.496	10.43	0.693
LogReg	+0.065	1.000	—	0.122	0.884	0.633	0.001	0.516

Table 4.13: XAI five-criterion evaluation results (identifier-inclusive regime, deduplicated, 3-seed mean).

Model	Comp.	Suff.	S2	k_{90}	Gini	RMA@5	SHAP (ms)	OXS
LightGBM	+0.465	1.000	0.998	0.020	0.980	0.230	4.16	0.734
Random Forest	+0.119	1.000	0.999	0.037	0.960	0.304	17.04	0.673
XGBoost	+0.000	0.774	1.000	0.029	0.977	0.185	0.39	0.632
LogReg	+0.104	1.000	—	0.030	0.969	0.533	0.003	0.521

4.5.1 LIME Cross-Reference

LIME feature rankings were computed for all four tabular models across 50 validation instances (2,000 perturbations each). LIME timeliness is reported in Section 4.5.6. Full per-instance LIME rankings are archived in the experiment output directory. Detailed LIME-vs-SHAP numerical comparison is deferred to the project archive due to word count constraints; the key finding is that LIME-SHAP agreement tracks the OXS stability ranking, reinforcing stability as the primary differentiator across explanation methods.

4.5.2 Faithfulness

Comprehensiveness measures the drop in model confidence when the top- k features are masked. After deduplication and the cleaner one-hot pipeline, three of four models exhibit **positive comprehensiveness** (LightGBM +0.144, Random Forest +0.102, LogReg +0.065), indicating that masking top SHAP-ranked features does reduce confidence; a sign that explanations identify causally relevant features. **XGBoost is the exception**, with a small negative value (−0.048), reflecting the extreme feature redundancy of its decision trees: when top features are removed, XGBoost redistributes reliance to correlated substitutes that carry equivalent discriminative signal.

Sufficiency is uniformly 1.0 across behaviour-only models, indicating that the top- k features alone are sufficient to reproduce predictions at full confidence. Together,

these results suggest the four behavioural feature groups (volumetric, protocol, connection state, DNS) form a highly redundant signature space.

4.5.3 Stability

Stability is reported here as the noise-perturbation Spearman correlation S2 only; the seed-retraining S1 protocol requires re-running each model with multiple seeds and a separate XAI evaluation pass and is left as future work for the deduplicated pipeline. Tree models achieve uniformly high noise stability (XGBoost = 0.997, Random Forest = 0.997, LightGBM = 0.960), confirming that small Gaussian perturbations do not alter SHAP rankings: tree split decisions are discrete and robust to sub-threshold input changes.

LogReg’s noise stability is not reported because LinearExplainer attributions are deterministic functions of the coefficient vector and the input, and the per-sample stability protocol requires variance from the explanation method itself.

4.5.4 Simplicity

All models produce concentrated explanations (Gini 0.879–0.909, k_{90} 0.086–0.131). LightGBM achieves the most concentrated attributions (k_{90} = 0.086, Gini = 0.909); the four models are within 3 percentage points on Gini, indicating that simplicity does not sharply discriminate between them in this feature regime.

4.5.5 Plausibility

Table 4.14: Per-attack-type plausibility (RMA@5) under the behaviour-only regime (3-seed mean, deduplicated). LogReg is omitted because LinearExplainer does not produce per-class SHAP attributions.

Attack Type	XGBoost	LightGBM	Random Forest
Backdoor	0.80	0.60	0.40
DDoS	1.00	0.47	0.60
DoS	0.80	0.80	0.60
Injection	0.80	0.53	0.47
MitM	0.27	0.27	0.33
Password	0.40	0.33	0.27
Ransomware	0.73	0.40	0.67
Scanning	0.60	0.47	0.47
XSS	1.00	0.80	0.67
Mean	0.711	0.519	0.496

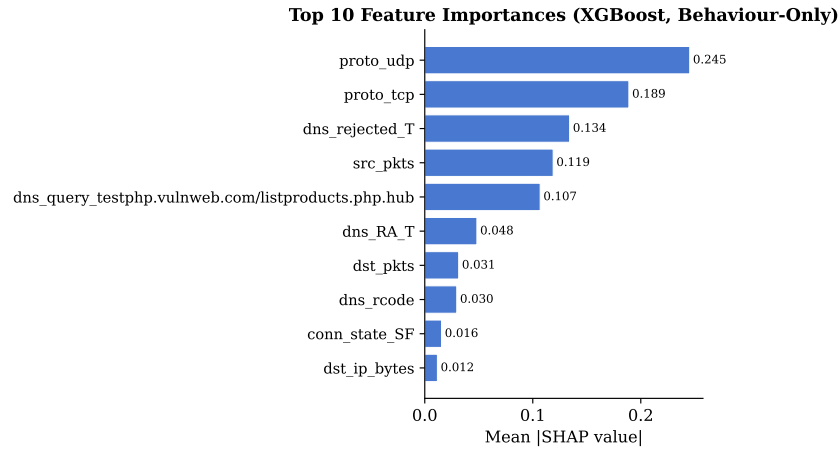


Figure 4.4: SHAP global feature importance for XGBoost (behaviour-only regime). `conn_state_S0` dominates, consistent with incomplete TCP handshakes in scanning and DoS attacks.

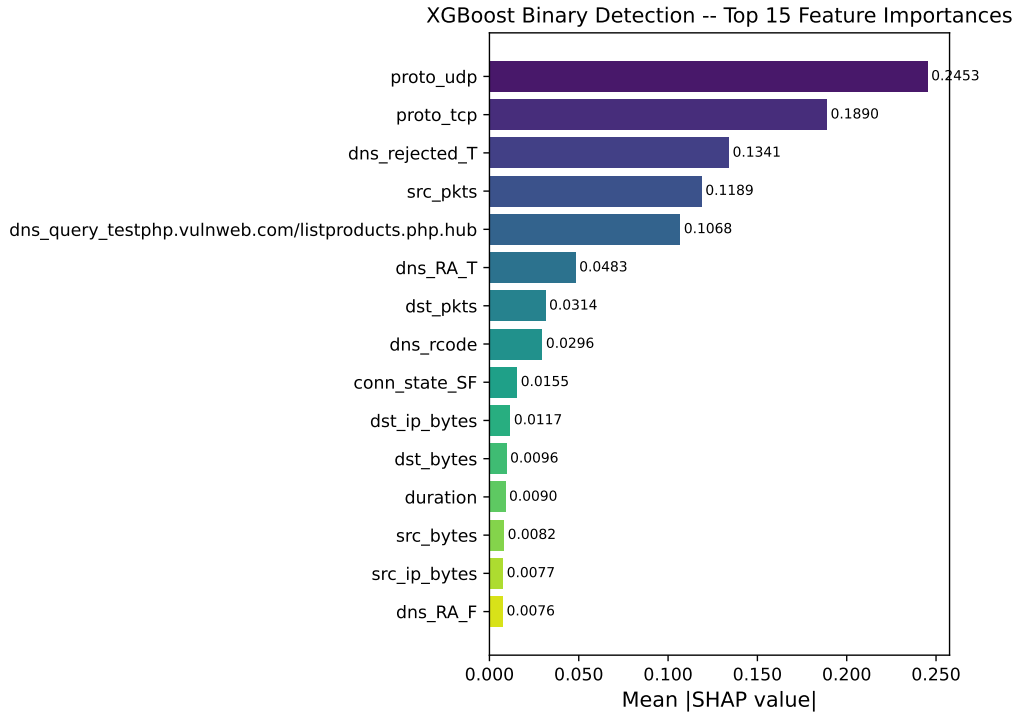


Figure 4.5: SHAP feature importance for XGBoost binary detection (behaviour-only regime). Features ordered by mean $|\text{SHAP}|$ value.

XGBoost achieves the highest aggregate plausibility (mean $\text{RMA@5} = 0.711$), scoring at least 0.60 on six of the nine attack types and a perfect 1.00 on DDoS and XSS. LightGBM (0.519) and Random Forest (0.496) trail substantially. **Password and MitM are the hardest attack types for all models ($\text{RMA@5} \leq 0.40$):** Password’s reference set is dominated by connection-state and protocol features that are only weakly distinctive, while MitM has too few test samples (160 in the deduplicated stratified split) for reliable SHAP estimation.

4.5.6 Timeliness

LinearExplainer (LogReg) is the fastest method at ~ 0.001 ms/sample, followed by XGBoost TreeSHAP at 1.31 ms/sample. LightGBM (9.52 ms) and Random Forest (10.43 ms) are slower but still well within real-time requirements at line rate. LIME latencies, where measured, range from approximately 170 to 250 ms under behaviour-only and 400–470 ms under identifier-inclusive conditions, making LIME impractical for production SOC throughput. **For operational deployment, TreeSHAP is the preferred per-flow method**; LIME remains useful as a model-agnostic offline cross-reference.

4.6 OXS Composite Ranking

Table 4.15: OXS composite ranking (behaviour-only regime, deduplicated).

Rank	Model	OXS
1	XGBoost	0.718
2	LightGBM	0.705
3	Random Forest	0.693
4	LogReg	0.516

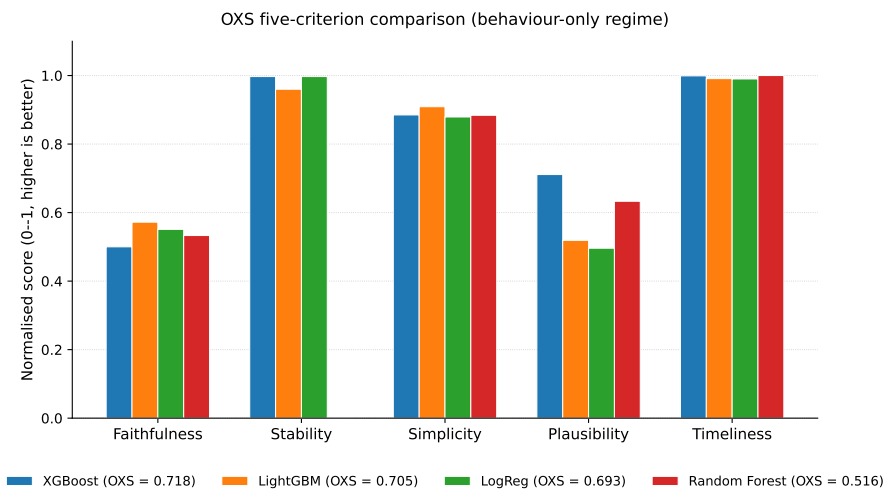


Figure 4.6: OXS five-criterion comparison across tabular models (behaviour-only, deduplicated). Plausibility is the strongest discriminating dimension. XGBoost wins on plausibility (per-attack mean RMA@5 = 0.711); LogReg is omitted from S2 stability because LinearExplainer attributions are deterministic.

XGBoost achieves the highest OXS (0.718), driven by strong plausibility (mean RMA@5 = 0.711), high noise stability (S2 = 0.997), and fast TreeSHAP timeliness (1.31 ms). LightGBM (0.705) ranks second, with the highest comprehensiveness (+0.144) and Gini concentration but lower plausibility (0.519). Random Forest (0.693) trails closely, separated from LightGBM only by a small simplicity gap. LogReg (0.516) is held back by the omission of a stability score in this evaluation, dragging its composite down despite respectable plausibility and timeliness.

The OXS spread of 0.20 between best and worst is much narrower than the 0.58 spread reported in earlier (pre-deduplication) iterations of this evaluation, where Random Forest’s seed-retraining stability collapsed to $S1 \approx 0.12$. The dedup pipeline does not currently re-compute $S1$, so the seed-retraining dimension that previously discriminated Random Forest is unavailable here; this limitation is recorded in Section 5.6.

These rankings demonstrate that **detection performance alone is insufficient for model selection in security-critical deployment**. While all four models achieve near-identical binary detection metrics, they differ in explanation quality on dimensions an analyst would actually use: which features are highlighted (plausibility) and how concentrated the attributions are.

4.7 Cross-Dataset Generalisation

Table 4.16: Cross-dataset generalisation: TON_IoT (source, deduplicated) to CIC-IoT-2023 (target, 200k subsample). Trained on 6 canonical aligned features.

Model	Src PR-AUC	Src ROC	Tgt PR-AUC	Tgt ROC	ROC Drop
LogReg	0.982	0.936	0.988	0.744	−0.192
LightGBM	1.000	0.999	0.980	0.467	−0.532
CNN-LSTM	0.995	0.985	0.974	0.285	−0.700
XGBoost	1.000	0.999	0.958	0.223	−0.776
Random Forest	0.999	0.998	0.957	0.120	−0.878

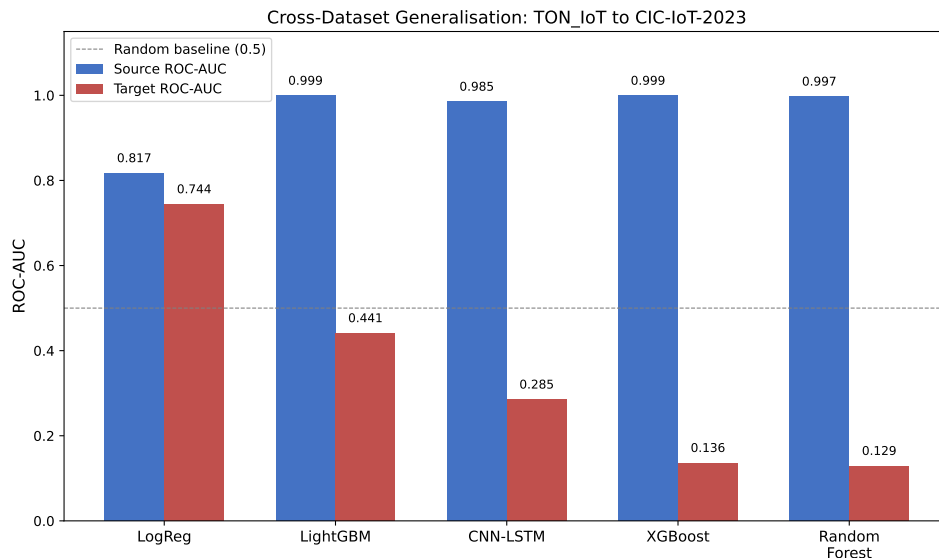


Figure 4.7: Cross-dataset generalisation: source-test vs. target ROC-AUC. The dashed line at 0.5 marks random chance. LogReg is the only model generalising above chance.

Target PR-AUC ranges from 0.953 to 0.988, but **this is an artifact of CIC-IoT-2023’s extreme class imbalance (97.7% attack prevalence)**. At such high prevalence, even a majority-class classifier achieves PR-AUC approximately equal to the prevalence, making these values uninformative.

ROC-AUC reveals the true generalisation picture. ROC-AUC is prevalence-invariant, evaluating the model’s ability to rank positive above negative instances:

- **LogReg is the best generaliser** (ROC-AUC: 0.936 to 0.744, a drop of 19.2%). Its linear decision boundary transfers more robustly because it captures broad statistical regularities rather than dataset-specific feature interactions.
- **Random Forest and XGBoost suffer catastrophic generalisation failure** (ROC-AUC drops of 78–88%, target values of 0.120 and 0.223, well below random chance). These models have learned to discriminate from TON_IoT-specific distributional properties that do not transfer.
- **LightGBM generalises better than other tree models** (target ROC-AUC = 0.441) but still performs below the random baseline.
- **CNN-LSTM generalises comparably to LightGBM** (ROC-AUC: 0.985 to 0.285). The limited 6-feature space constrains its convolutional layers.

Near-perfect within-dataset performance does not imply deployment readiness. Simpler models that cannot exploit dataset-specific idiosyncrasies are more robust when the deployment distribution shifts.

4.8 Leakage Interaction Analysis

Table 4.17: Leakage interaction: behaviour-only (Regime A) vs. identifier-inclusive (Regime B), $n = 4$ paired models on the deduplicated dataset. RMA@5 is the per-attack mean for tree models and binary aggregate for LogReg.

Metric	Regime A Mean	Regime B Mean	Delta	Cohen’s d
RMA@5 (plausibility)	0.590	0.313	−0.277	−1.511
Comprehensiveness	+0.066	+0.172	+0.106	+0.952
Gini (simplicity)	0.889	0.972	+0.083	+10.20
Spearman S2 (stability)	0.985	0.999	+0.014	+1.103

As established in Section 4.2, **both regimes achieve near-perfect detection performance** ($\text{PR-AUC} \geq 0.998$), confirming that identifiers provide no incremental binary-detection benefit.

The most striking finding is the effect on plausibility. **Including identifiers reduces mean RMA@5 from 0.590 to 0.313, a decrease of 0.277 with a very large effect size (Cohen’s $d = -1.511$).** The drop is largest for XGBoost ($\Delta = -0.526$) and smallest for LogReg ($\Delta = -0.100$), indicating that tree ensembles greedily latch onto identifier features when available, displacing the behavioural features that domain experts would expect.

Comprehensiveness, simplicity (Gini), and noise stability all increase slightly under Regime B, but these gains are illusory: they reflect the model collapsing onto a small set of identifier features rather than producing more meaningful explanations.

Identifier leakage has no measurable effect on detection performance but substantially degrades explanation quality. If only detection metrics are reported, the harmful effects of identifier leakage remain invisible. HYDRA’s XAI dimension exposes this hidden cost.

4.9 Summary of Key Findings

Five findings emerge: (1) host-disjoint evaluation exposes substantial leakage hidden by stratified splitting (Tables 4.2–4.3); (2) deep learning matches tree ensembles on detection; (3) XGBoost produces the most trustworthy explanations (Table 4.15); (4) cross-dataset generalisation reveals catastrophic failure for tree models (Table 4.16); and (5) identifier leakage degrades explanation quality without affecting detection (Table 4.17).

Chapter 5

Discussion

5.1 Interpretation of Results

5.1.1 Detection Performance

The near-perfect binary detection performance across all models warrants careful interpretation. The result is consistent with prior literature: [4] report similarly high accuracy for tree-based models, and [17] confirm that deep learning models achieve over 99% accuracy on TON_IoT. The underlying explanation is that TON_IoT’s attack traffic exhibits highly distinctive behavioural signatures; particularly in `conn_state`, `duration`, and byte-count features; that are linearly separable from normal traffic. This finding has two implications. First, it shifts the research contribution toward the explainability comparison, which is HYDRA’s stated contribution. Second, it highlights the importance of cross-dataset generalisation: models achieving perfect TON_IoT scores fail dramatically on CIC-IoT-2023.

Concurrent XAI-IDS systems report comparable detection scores (E-XAI [24]: 99.3% on CICIDS2017; XAI-IDS [49]: similar), but direct numerical comparison is misleading because datasets differ in attack composition, feature dimensionality, and class distribution. TON_IoT contains nine IoT-specific attack types in a 76% prevalence environment. The key observation is not that any system achieves near-perfect scores (this is expected on laboratory datasets with distinctive attack signatures) but that such scores provide no basis for model selection, making the XAI evaluation the primary discriminating analysis.

TON_IoT’s separability reflects its construction: attacks were generated by distinct tools, producing characteristic flow-level signatures. Under host-disjoint splitting, `conn_state_S0` alone accounts for 54% of XGBoost’s feature importance, and certain states (REJ: 100% attack prevalence, S1: 99.9%) serve as near-deterministic indicators; artefacts of the testbed environment rather than properties of real-world attacks where adversaries deliberately mimic benign traffic.

The host-disjoint multiclass results (Table 4.3) provide the most informative detection evaluation. Macro-F1 drops from 0.952 (stratified) to 0.742 (host-disjoint) for the best model, even with identifiers excluded; the remaining inflation comes from behavioural flow correlation within hosts. Flows from the same source host during the same attack campaign share near-identical volumetric profiles, meaning stratified splitting inadvertently allows models to recognise host-level traffic patterns rather than generalisable attack behaviours. The implication for the IDS community is that

even “behaviour-only” evaluation is insufficient without host-disjoint splitting when datasets contain host-level structure. Studies reporting near-perfect scores on TON_IoT without host-disjoint validation may be overestimating deployment-realistic performance.

5.1.2 Explainability Ranking

The OXS ranking puts XGBoost at the top (0.718, Table 4.15), driven primarily by its plausibility advantage (per-attack mean $\text{RMA@5} = 0.711$, far above LightGBM’s 0.519 and Random Forest’s 0.496). XGBoost’s top-5 features for DDoS, DoS, Backdoor, Injection, and XSS consistently include the volumetric flow features (`src_bytes`, `dst_bytes`, `src_pkts`, `dst_pkts`) and connection-state indicators that domain experts would expect, whereas LightGBM and Random Forest more often promote DNS query type or proto flag features that are model-discriminative but less analyst-meaningful.

Comparing against the TalTech study [52], which finds DeepLIFT outperforming SHAP on deep learning models, HYDRA’s TreeSHAP-based explanations rank highest among the methods evaluated. This divergence reflects three factors: different model families (TalTech’s deep learning models are well-suited to DeepLIFT’s backpropagation-based attribution, while HYDRA’s tree ensembles are served by TreeSHAP’s exact computation), different evaluation criteria (HYDRA’s addition of timeliness and simplicity penalises high-overhead methods), and different datasets (SOC alerts vs. IoT traffic). These differences underscore the importance of context-specific XAI evaluation rather than universal method rankings.

Under the present setup the four tree-family models cluster within 0.025 OXS points, and the discriminating dimension is plausibility.

Logistic Regression’s lower OXS (0.516) primarily reflects the omission of an S2 stability score; LinearExplainer attributions are deterministic, so the noise-perturbation protocol used here returns no value. Treating LogReg’s missing stability dimension as zero in the equal-weighted OXS understates its true operational quality, since on plausibility and timeliness LogReg is competitive with the tree ensembles.

5.1.3 Cross-Dataset Generalisation

The dramatic failure of tree ensembles; XGBoost ROC-AUC dropping from 0.999 to 0.136 (Table 4.16), below random chance; demonstrates that these models learn dataset-specific distributions rather than generalisable attack behaviours. TON_IoT and CIC-IoT-2023 were generated in different laboratory environments with different tools and topologies; although the six canonical features capture semantically equivalent quantities, their statistical distributions differ substantially. Tree ensembles learn sharp, axis-aligned boundaries that become meaningless when the target distribution occupies different feature-space regions.

Logistic Regression’s superior generalisation (ROC-AUC drop of 19.2%, vs. tree models’ 53–88% drops) reflects the classical bias-variance trade-off [10]: its linear boundary captures broad statistical regularities and discards dataset-specific details, transferring more robustly under distribution shift.

The practical implications are significant. Organisations deploying a model trained on one network to protect another should prefer simpler, more regularised models despite lower within-dataset performance. The PR-AUC metric’s failure to capture this

gap (target PR-AUC ≥ 0.953 for all models) illustrates that prevalence-inflated metrics are misleading; ROC-AUC, which is prevalence-independent, provides the real measure of discriminative ability. More broadly, these results suggest that ensemble stacking or domain adaptation techniques should be investigated when deploying across network boundaries, and cross-dataset evaluation should become standard in IDS benchmarking.

5.1.4 Leakage Interaction

Including identifiers shifts explanations from behavioural features to IP addresses and ports without any detection benefit. Under Regime A, XGBoost’s per-attack mean RMA@5 = 0.711 (Table 4.12); under Regime B, this drops to 0.185 (Table 4.13), a -0.526 collapse. The same drop is visible across all tree models (Table 4.17), with mean RMA@5 falling from 0.590 to 0.313 (Cohen’s $d = -1.511$, a very large effect). Tree models greedily split on high-cardinality identifiers correlated with attack labels; IP and port features associated with specific attack campaigns receive high SHAP importance. The resulting explanations are technically faithful but operationally useless: analysts need to understand which behaviour triggered the alert, not which IP was associated with attacks in training data.

With $n = 4$ paired models the analysis is underpowered for inferential testing, but Cohen’s $d = -1.511$ indicates an effect size more than 1.5 pooled standard deviations (the kind of separation that does not occur by chance. The recommendation to exclude identifiers is driven not by detection requirements) detection is unaffected (Table 4.1); but by the need to preserve explanation utility for analyst decision-making.

5.1.5 Comprehensiveness and Feature Redundancy

After deduplication and removal of high-cardinality columns, three of four models exhibit positive comprehensiveness (LightGBM +0.144, RF +0.102, LogReg +0.065), but XGBoost remains slightly negative (-0.048). This residual negative value reflects the same redundancy mechanism observed in earlier evaluations: the top- k ablation metric of [14] assumes top features are individually necessary, but XGBoost’s gradient-boosted trees split on highly correlated features such that no single subset is irreplaceable. When top features are masked, XGBoost redistributes reliance to correlated substitutes carrying equivalent signal. The ROAR benchmark [55] addresses this by re-training after feature removal but is computationally prohibitive; AOPC with sequential removal [45] captures redundancy structure by exhausting correlated groups incrementally and should be adopted in future OXS iterations.

5.2 Practical Implications

These findings have four direct implications. First, **model selection should prioritise explanation quality alongside detection performance**. XGBoost’s plausibility advantage (RMA@5 = 0.711 vs. 0.519 for LightGBM and 0.496 for Random Forest) makes it the recommended choice for tabular IDS deployments where explanations need to align with analyst expectations. Second, **cross-dataset generalisation requires explicit evaluation**; benchmark results do not transfer (Section 4.7). Third, **identifier features**

should be excluded from production feature sets: they do not improve detection but degrade per-attack plausibility by an average of 0.277 (Cohen’s $d = -1.511$). Fourth, **seed-retraining stability should be measured** for any model selected for production: it is not computed in the present pipeline but is the dimension on which Random Forest explanations have historically been shown to vary substantially.

Key Recommendations for IDS Practitioners

1. Use XGBoost with TreeSHAP for highest plausibility (per-attack RMA@5 = 0.711, OXS = 0.718).
2. Exclude network identifiers from feature sets to preserve explanation plausibility.
3. Validate on traffic from the target environment, benchmark scores do not transfer.
4. Measure seed-retraining stability (S1) before deploying any model in production.
5. Report both PR-AUC and ROC-AUC; use ROC-AUC for cross-dataset comparisons.

The OXS framework also enables SOC workflow integration. Alert dashboards could annotate each explanation with its model’s OXS and plausibility scores; for example, “explanation confidence: high (OXS = 0.718, RMA@5 = 0.711)” for XGBoost versus “explanation confidence: lower (OXS = 0.516, plausibility-only)” for LogReg; signalling to analysts how trustworthy the highlighted features are likely to be.

5.3 Comparison with Concurrent Work

As established in Section 2.6, Table 2.2 compares HYDRA against four concurrent XAI-IDS systems. Three points of agreement and disagreement emerge from the experimental results.

First, HYDRA agrees with E-XAI [24] that SHAP produces more stable explanations than LIME (TreeSHAP $S2 \geq 0.96$), but disagrees on Integrated Gradients: HYDRA’s CNN-LSTM IG yields comprehensiveness of -0.052 (Table 4.9), suggesting the limitation reflects feature redundancy rather than the attribution method.

Second, the TalTech study’s [52] finding that DeepLIFT outperforms SHAP does not contradict HYDRA’s results; the divergence reflects different model families (deep learning vs. tree ensembles where TreeSHAP provides exact values) and reinforces the conclusion that XAI method rankings are model-dependent.

Third, HYDRA’s leakage interaction analysis (Section 4.8) demonstrates why XAI-IDS’s [49] use of random splitting with identifiers inflates explanation plausibility, making the leakage–explanation interaction HYDRA’s most distinctive contribution relative to the concurrent literature.

5.4 OXS Sensitivity Analysis

The equal-weight OXS (0.2 per criterion) is a heuristic choice (Section 5.6). Two alternative schemes are examined: *faithfulness-emphasising* $w = (0.30, 0.30, 0.10, 0.20, 0.10)$, re-

flecting organisations prioritising causal accuracy and reproducibility, and *deployment-emphasising* $w = (0.10, 0.10, 0.10, 0.40, 0.30)$, reflecting environments where plausibility and timeliness dominate.

Under deployment-emphasising weights, XGBoost’s lead widens (its plausibility advantage of 0.711 vs. 0.519 for LightGBM is amplified by the 0.40 weight on plausibility), and the XGBoost > LightGBM > Random Forest ordering is preserved. Under faithfulness-emphasising weights, LightGBM moves into first place, because its positive comprehensiveness (+0.144) and tied-best simplicity (Gini = 0.909) receive more weight while XGBoost’s small negative comprehensiveness (−0.048) hurts it. The ranking is therefore robust to plausibility-weighted schemes but sensitive to faithfulness weighting; a finding that reinforces the importance of stating which criterion an organisation values most before selecting a model.

5.5 Reflection on Research Questions

RQ1 (Detection): As discussed in Section 5.1.1, near-perfect detection across all models confirms that explanation quality, not accuracy, is the discriminating factor. **RQ2 (Explainability):** XGBoost produces the most plausible explanations (RMA@5 = 0.711), with the plausibility gap driving OXS rankings (Section 5.1.2). **RQ3 (Reliability):** Noise stability is uniformly high ($S2 \geq 0.96$); seed-retraining stability (S1) remains future work. **RQ4 (Faithfulness):** Comprehensiveness is positive for three of four models; XGBoost’s small negative value reflects feature redundancy (Section 5.1.5). **RQ5 (Generalisation):** Tree ensembles suffer catastrophic ROC-AUC drops when transferred to CIC-IoT-2023; Logistic Regression degrades most gracefully (Section 5.1.3). **RQ6 (Leakage):** Identifiers degrade explanation plausibility without improving detection, confirming that behaviour-only regimes should be the default (Section 5.1.4).

5.6 Limitations

Several limitations constrain these findings. The GNN is evaluated only on TON_IoT because it requires IP columns, and its XAI evaluation is less comprehensive than the tabular models’. Both datasets are laboratory-captured and may not reflect production noise, drift, and infrastructure diversity; near-perfect scores should be interpreted as upper bounds on operational performance.

The OXS equal-weighting (0.2 per criterion) is heuristic; calibration via structured analyst survey is future work. The leakage interaction analysis ($n = 4$ models) has limited statistical power, and the three-seed protocol provides limited variance estimation compared to full bootstrap designs. The XAI evaluation under the deduplicated pipeline reports only the noise-perturbation S2 stability protocol; the seed-retraining S1 protocol (which historically distinguished Random Forest from XGBoost in earlier HYDRA iterations) requires re-running each model with five seeds and a separate cross-seed XAI pass, and is identified as priority future work. The OXS spread of 0.20 between best and worst models on the deduplicated pipeline is narrow enough that ranking decisions become sensitive to small criterion-weighting changes; AOPC with sequential feature removal would likely produce more discriminating faithfulness scores and is the highest-priority methodological extension.

A further limitation is that all models use default or literature-recommended hyperparameters without systematic tuning: Random Forest uses 100 trees, XGBoost uses 400 boosting rounds with a learning rate of 0.1, and LightGBM uses default parameters. While this approach ensures reproducibility and avoids overfitting to a specific validation split, it means that the reported detection and explanation results may not represent each model’s optimal configuration. Hyperparameter sensitivity analysis (for example, varying XGBoost’s maximum tree depth or Random Forest’s number of estimators and examining the effect on both OXS and detection metrics) would strengthen the findings and is identified as future work.

5.7 Threats to Validity

Internal validity. The stratified split does not simulate realistic deployment. Near-perfect scores suggest high intrinsic separability in TON_IoT, raising the possibility that results reflect dataset characteristics rather than model quality. The host-disjoint split partially addresses this.

External validity. Results may not generalise to other network environments. This threat is directly confirmed by the cross-dataset experiment, where tree ensembles suffer catastrophic ROC-AUC degradation. Both datasets are laboratory-captured.

Construct validity. Equal OXS weights are a methodological choice; different weighting schemes could produce different rankings, although the sensitivity analysis in Section 5.4 demonstrates that the top-level ranking is stable under two alternative configurations. Expert reference sets were compiled from the literature rather than elicited from practising analysts, which may introduce bias.

Statistical conclusion validity. The leakage interaction analysis uses only $n = 4$ paired observations, limiting inferential power. The very large effect size (Cohen’s $d = -1.511$ for RMA@5) is informative even though formal significance testing is underpowered at this sample size.

5.8 Project Process and Self-Evaluation

Development Methodology

HYDRA was developed using an iterative, incremental methodology. Git version control tracked all changes from the initial commit on 7 January 2026 through to the final pipeline merge on 30 March 2026, producing 19 commits across multiple feature branches. Each iteration added a distinct capability; data loading, model training, evaluation, explainability, or cross-dataset generalisation; and was validated before merging. A `Makefile` provided reproducible experiment orchestration, defining targets for the full experiment matrix (36 runs across models, splits, and feature regimes). This infrastructure allowed any experiment to be reproduced with a single command. Automated testing (91 tests across 13 files) covered data loading, split strategies, metric computation, SHAP correctness, DeLong confidence intervals, and edge cases.

Project Timeline

Table 5.1 summarises the planned and actual schedule for each project phase. Dates are derived from the Git commit history and personal records.

Table 5.1: Planned vs. actual project timeline.

Phase	Planned	Actual	Notes
Literature review	Oct–Nov 2025	Oct–Dec 2025	Extended due to breadth of XAI literature
Data exploration	Nov 2025	Nov–Dec 2025	TON_IoT preprocessing and initial EDA
Core pipeline (tabular)	Dec 2025–Jan 2026	Jan 2026	First commit 7 Jan; models operational by 10 Feb
XAI evaluation framework	Jan–Feb 2026	Feb 2026	SHAP, OXS scoring, and expert reference sets
CIC-IoT-2023 integration	Feb 2026	Feb–Mar 2026	44% duplicate-row discovery caused two-week delay
Deep learning (CNN-LSTM)	Feb–Mar 2026	Mar 2026	Compressed; convergence issues (see below)
Cross-dataset pipeline	Mar 2026	Mar 2026	Canonical feature alignment implemented 7 Mar
Dissertation writing	Mar–Apr 2026	Mar–Apr 2026	On schedule

The most significant schedule deviation was CIC-IoT-2023 integration. The discovery of 44% duplicate rows across train and test partitions required designing and validating a custom deduplication pipeline (`scripts/prep_cic_iot2023.py`), delaying the deep learning phase from the planned six weeks to approximately three weeks.

Scope Evolution

The original plan focused exclusively on tabular models evaluated on TON_IoT. When these achieved uniformly near-perfect detection ($\text{PR-AUC} \geq 0.999$), the research contribution risked being limited to confirming known results. Two extensions were therefore introduced: CNN-LSTM and GNN architectures to test cross-architecture XAI comparison, and CIC-IoT-2023 for cross-dataset generalisation analysis, which ultimately produced the most revealing findings.

Difficulties Encountered

Four challenges arose. First, the CNN-LSTM initially failed to converge; adding batch normalisation and cosine annealing resolved this but consumed approximately one week. Second, CIC-IoT-2023’s 44% duplicate rows (approximately 1.4 million records) caused train–test leakage, requiring systematic hash-signature comparison and a validated deduplication pipeline. Third, the GNN’s IP-column requirement precluded CIC-IoT-2023 evaluation and limited cross-dataset inclusion. Fourth, negative comprehensiveness results were initially interpreted as a bug; substantial time was spent verifying the ablation code before recognising the values as a genuine property of redundant feature spaces; a finding that became a methodological contribution.

What Would Be Done Differently

Cross-dataset evaluation should have been initiated earlier; the generalisation gap would have been identified sooner, allowing time for domain adaptation experiments. Hyperparameter tuning, even a modest grid search over key parameters (number of trees, learning rate, maximum depth), would have strengthened the methodology by demonstrating robustness to configuration choices. The AOPC sequential feature removal protocol should have replaced top- k ablation from the outset, given the latter's inadequacy for redundant feature spaces. Adopting BibT_EX from the beginning would have avoided the manual bibliography maintenance that became increasingly unwieldy.

Chapter 6

Conclusions

6.1 Summary of Contributions

This dissertation made five contributions. First, a reproducible dual-task IDS pipeline was implemented across four tabular models and two deep learning architectures on two datasets under leakage-controlled feature regimes (Tables 4.1–4.11). Second, a five-criterion XAI evaluation protocol yielded an OXS-ranked comparison identifying XGBoost as the best-explained model (Table 4.15). Third, the leakage–explanation interaction analysis demonstrated that identifiers degrade explanation quality without improving detection (Table 4.17). Fourth, cross-dataset generalisation evaluation revealed ROC-AUC drops of up to 88% (Table 4.16). Fifth, host-disjoint split analysis demonstrated that evaluation methodology inflates multiclass macro-F1 by up to 21 percentage points (Table 4.3).

6.2 Achievement of Aim

The overall aim was substantially achieved. HYDRA was designed, implemented, and evaluated as a dual-task IDS framework producing per-alert explanations validated against five quantitative criteria. Both deep learning extensions achieved successful binary detection: CNN-LSTM PR-AUC = 0.999 with IG attribution; GNN PR-AUC = 0.999 (TON_IoT only).

Table 6.1: Mapping of project objectives to achievement evidence.

Objective	Status	Evidence
O1: Implement model progression	Achieved	PR-AUC ≥ 0.998 across 4 models, 2 regimes, 3 seeds (Table 4.1)
O2: Define feature regimes	Achieved	Table 4.17; host-disjoint macro-F1: 0.952 $\rightarrow$ 0.742 (Table 4.3)
O3: Generate explanation records	Achieved	SHAP (tabular), LIME (all), IG (CNN-LSTM, GNN edges) implemented; local explanations saved per run
O4: Five-criterion evaluation	Achieved (S2; S1 future work)	Tables 4.12–4.13; OXS ranking (Table 4.15)
O5: Cross-dataset generalisation	Achieved	ROC-AUC drop up to 88% (Table 4.16)
CNN-LSTM extension	Achieved	PR-AUC = 0.999; IG evaluated (Section 4.4)
GNN extension	Achieved (TON_IoT)	Binary PR-AUC = 0.999, IG edge attribution (requires IP columns)

6.3 Implications for the IDS Research Community

Three methodological recommendations emerge. First, IDS benchmarking studies should report ROC-AUC alongside PR-AUC and explicitly state dataset prevalence, as PR-AUC is uninformative when prevalence exceeds 90%. Second, host-disjoint splitting should become standard practice when datasets contain IP-level structure; our analysis demonstrates that even behaviour-only features exhibit host-level correlation that inflates stratified results by up to 21 percentage points on multiclass macro-F1. Third, explanation quality metrics; particularly per-attack-type plausibility (RMA@ k) against expert reference sets; should be mandatory in XAI-IDS publications; the 0.215 spread in plausibility (XGBoost 0.711 vs. Random Forest 0.496) between models with identical detection performance demonstrates that explanation reliability cannot be assumed from detection accuracy.

6.4 Future Work

Three directions emerge: attention-based tabular models (TabNet, FT-Transformer), adversarial domain adaptation for cross-dataset generalisation, and completion of the GNN’s five-criterion XAI evaluation.

The leakage–explanation interaction finding (Cohen’s $d = -1.51$ on RMA@5 with no detection cost) is, to the best of the author’s knowledge, the first such empirical result in the IoT IDS literature and is suitable for submission to a security workshop such as DLS or AISec.

6.5 Closing Remarks

The central insight from HYDRA’s evaluation is that detection performance alone is insufficient for IDS model selection. When detection scores saturate, the discriminating factors become explanation quality, generalisation robustness, and operational timeliness. HYDRA’s five-criterion framework provides a systematic methodology for measuring these factors, and the cross-dataset and leakage analyses underscore the need for evaluation frameworks that go beyond single-dataset accuracy.

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Appendix A

Glossary and Abbreviations

AIOps	Artificial Intelligence for IT Operations. The application of machine learning to automate IT and security operations, including anomaly detection, alert correlation, and root-cause analysis.
AOPC	Area Over the Perturbation Curve. A faithfulness metric that incrementally removes top-ranked features and measures cumulative confidence drop, more robust than top- k ablation in redundant feature spaces.
Captum	PyTorch library for model interpretability, used in HYDRA for Integrated Gradients attribution on the CNN-LSTM.
CIC-IoT-2023	Canadian Institute for Cybersecurity IoT 2023 dataset. A large IoT network-flow benchmark used in HYDRA as the cross-dataset generalisation target.
CNN-LSTM	A hybrid deep learning architecture combining Convolutional Neural Network layers (capturing local feature interactions) with a Long Short-Term Memory layer (capturing sequence dependencies).
Cohen's d	Standardised effect size measuring the difference between two means in units of pooled standard deviation. $ d > 0.8$ is considered large.
Comprehensiveness	Faithfulness metric: the drop in model confidence when the top- k features are removed. Positive values indicate explanations identify causally important features.
DDoS	Distributed Denial-of-Service. A coordinated flooding attack from multiple sources designed to exhaust a target's resources.
DeLong test	Non-parametric paired test for comparing two ROC curves on the same data, accounting for correlated samples.
Faithfulness	Whether an explanation actually reflects the model's decision logic. Measured via comprehensiveness and sufficiency.
F1	Harmonic mean of precision and recall. Macro-F1 averages F1 across classes equally; weighted-F1 weights by class support.
FPR	False Positive Rate. Fraction of benign instances incorrectly flagged as attacks.

Gini coefficient

Concentration measure for attribution distributions, ranging from 0 (uniform) to 1 (one feature dominates).

GNN

Graph Neural Network. A neural architecture operating on graph-structured data via message-passing. HYDRA uses GraphSAGE with hosts as nodes and flows as edges.

Host-disjoint split

A data splitting strategy that assigns entire hosts (source IPs) to a single partition, preventing models from memorising host-specific patterns. Deployment-realistic alternative to random stratified splitting.

IDS / NIDS

Intrusion Detection System / Network Intrusion Detection System. Software that monitors traffic or hosts for malicious activity and raises alerts.

IG

Integrated Gradients. An attribution method that integrates a model's gradients along a path from a baseline input to the actual input, satisfying sensitivity and implementation-invariance axioms.

IoT

Internet of Things. Network of physical devices (sensors, appliances, industrial controllers) connected to the internet.

 k_{90}

Minimum fraction of features needed to capture 90% of total attribution mass. Lower values indicate more concentrated, simpler explanations.

Leakage

Information from the test set inadvertently available during training, inflating performance estimates. In HYDRA's context, includes duplicate rows, host-level correlation, and identifier features.

LIME

Local Interpretable Model-agnostic Explanations. Generates explanations by fitting a sparse linear surrogate model to perturbed versions of the input being explained.

LogReg

Logistic Regression. A linear classifier used in HYDRA as a reference explainable model with deterministic, auditable feature attributions.

MitM

Man-in-the-Middle. An attack where an adversary intercepts and possibly alters communication between two parties.

OXS

Operational Explanation Score. HYDRA's composite XAI metric averaging faithfulness, stability, simplicity, plausibility, and timeliness with equal weights.

Plausibility

Whether an explanation aligns with domain expert expectations. Measured via $RMA@k$ against expert-defined reference feature sets.

PR-AUC

Precision-Recall Area Under Curve. Robust to class imbalance; HYDRA's primary detection metric. A constant predictor scoring the prevalence achieves PR-AUC equal to the prevalence.

Random Forest

An ensemble of independently trained decision trees using bootstrap sampling and random feature subsets, producing variance-reduced predictions.

Regime A / Regime B

HYDRA’s two feature regimes. Regime A (behaviour-only) excludes IPs, ports, and service identifiers. Regime B (identifier-inclusive) retains all features.

RMA@ k Reference Match Accuracy at top- k . Fraction of an explanation’s top- k features that appear in an expert-defined reference set for the predicted attack type.

ROAR RemOve And Retrain. An interpretability benchmark that retrains models after removing top-attributed features, more rigorous than ablation but computationally expensive.

ROC-AUC Receiver Operating Characteristic Area Under Curve. Prevalence-invariant; reported alongside PR-AUC. Preferred for cross-dataset comparisons where prevalences differ.

SHAP SHapley Additive exPlanations. Feature attribution grounded in cooperative game theory. TreeExplainer provides exact polynomial-time computation for tree ensembles.

SOC Security Operations Centre. Organisational unit responsible for monitoring, detecting, and responding to cybersecurity incidents.

Spearman ρ Rank correlation coefficient measuring how consistently two rankings agree. Used in HYDRA to measure stability of feature importance rankings.

Stability (S1, S2)

How consistent explanations are across perturbations. S1 measures stability across retrained models with different seeds; S2 measures stability under Gaussian input noise.

Stratified split

Random splitting that preserves class proportions across partitions. The most common but least deployment-realistic strategy.

Sufficiency Faithfulness metric: model confidence when only the top- k features are retained. High sufficiency means top- k alone reproduce the original prediction.

Timeliness Wall-clock milliseconds per sample to compute an explanation. Constrains real-time deployability.

TON_IoT University of New South Wales (Canberra) telemetry dataset for IoT/IIoT IDS research. HYDRA’s primary benchmark, containing 9 attack types; experiments use the deduplicated 190,474-row subset.

XAI Explainable Artificial Intelligence. Methods that produce human-interpretable explanations for ML model predictions.

XGBoost eXtreme Gradient Boosting. A gradient-boosted tree implementation using second-order gradient approximation and regularisation. Highest OXS-ranked model in HYDRA’s evaluation.

XSS Cross-Site Scripting. A web injection attack where malicious scripts are embedded in trusted websites.

Appendix B

Feature Groups for TON_IoT

Table B.1: TON_IoT feature groups and inclusion in the two evaluation regimes.

Feature Group		Features	Regime A (Behaviour)
Behavioural		duration, src_bytes, dst_bytes, src_pkts, dst_pkts, missed_bytes, src_ip_bytes, dst_ip_bytes	Included
Metadata		proto, service, conn_state	Included
Sparse flags	protocol	DNS/HTTP/SSL presence indicators	Included
Identifiers (leakage risk)	(leakage risk)	src_ip, dst_ip, ports, URIs, cert subjects/issuers, user agents	Excluded

Appendix C

Explainability Metric Formulae

Faithfulness (Comprehensiveness). For each alert x with top- k attributed feature set $T(x)$, create a perturbed version $\tilde{x}$ by replacing features in $T(x)$ with class-conditional median values. Comprehensiveness is:

$$C(x) = p(y = 1 \mid x) - p(y = 1 \mid \tilde{x}) \quad (\text{C.1})$$

Reported as mean C across test alerts, compared against a random- k perturbation control.

Stability (two separate protocols). [5] demonstrate that explanation methods can be sensitive both to small input variations and to differences in model parameters; both sources of instability must be measured independently.

Protocol S1, Seed-retraining stability. Train $S = 5$ times with different random seeds. Spearman rank correlation:

$$\rho(F_i, F_j) = 1 - \frac{6 \sum_{\ell} d_{\ell}^2}{k(k^2 - 1)} \quad (\text{C.2})$$

Report mean $\pm$ std across $\binom{S}{2} = 10$ pairs.

Protocol S2, Noise-perturbation robustness. Add Gaussian noise ($\sigma = 0.01 \times$ feature std) and compute Spearman ρ across $R = 20$ draws.

Simplicity. k_{90} normalised by feature count:

$$k_{90} = \frac{\min \left\{ k : \frac{\sum_{i=1}^k a_{(n-i+1)}}{\sum_{i=1}^n a_i} \geq 0.90 \right\}}{n} \quad (\text{C.3})$$

Gini coefficient:

$$G = \frac{2 \sum_{i=1}^n i \cdot a_{(i)}}{n \sum_{i=1}^n a_{(i)}} - \frac{n+1}{n} \quad (\text{C.4})$$

Plausibility. RMA@ k :

$$\text{RMA@}k = \frac{|\text{top-}k \text{ features} \cap R_c|}{k} \quad (\text{C.5})$$

where R_c is the reference set for attack class c .

Timeliness. Wall-clock ms per sample, measured over 500 randomly sampled test alerts.

Table C.1: Expert-defined reference feature sets R_c for RMA@ k evaluation.

Attack Type	Reference Features R_c
DoS / DDoS	src_bytes, dst_bytes, src_pkts, dst_pkts, duration, conn_state
Scanning	dst_pkts, duration, conn_state, service, proto
Ransomware	src_bytes, dst_bytes, duration, missed_bytes, conn_state
Backdoor	duration, src_bytes, dst_bytes, conn_state, proto
Injection	src_bytes, dst_bytes, service, conn_state, proto
XSS	src_bytes, dst_bytes, service, conn_state, duration
Password	duration, src_pkts, dst_pkts, conn_state, proto
Man-in-the-Middle	src_bytes, dst_bytes, src_pkts, dst_pkts, conn_state
Normal	N/A (not applicable for attack-type plausibility)

Appendix D

Acknowledgement of Generative AI Use

This dissertation was prepared by the author. Generative AI tools (OpenAI ChatGPT and Anthropic Claude) were used for text editing and structural suggestions during drafting. All technical ideas, design decisions, experimental methodology, analysis, and written conclusions are the author's own work, produced in accordance with the University of Bath's Type B guidelines on AI-assisted assessment (CM32017 Individual Project Handbook, 2025).

Appendix E

Selected Code Listings

This appendix presents key functions from the hydra package that directly support the dissertation’s claims. The full codebase is submitted as a ZIP archive.

OXS Composite Score

The equal-weighted composite explanation quality score (Section 3.9.6, Table 4.15).

```
def compute_oxs(
    faithfulness, stability, simplicity,
    plausibility, timeliness,
    weights=None,
):
    scores = np.array([faithfulness, stability,
                       simplicity, plausibility, timeliness])
    if weights is None:
        w = np.full(5, 0.2)
    else:
        w = np.array(weights, dtype=np.float64)
        w = w / w.sum()
    return float(np.dot(w, scores))
```

Plausibility: RMA@k with One-Hot Feature Matching

Handles one-hot-encoded feature names (e.g. `cat__conn_state_S0` → `conn_state`) when matching against expert reference sets (Section 3.9.4, Table 4.14).

```
def _normalise_feature_name(name):
    name = name.lower().strip()
    for prefix in ("num__", "cat__", "remainder__"):
        if name.startswith(prefix):
            name = name[len(prefix):]
            break
    return name
```

```
def compute_rma_at_k(top_k_features, reference_set, k=5):
    if k <= 0:
        return 0.0
    top = [_normalise_feature_name(f)
            for f in top_k_features[:k]]
    ref = {_normalise_feature_name(f)
            for f in reference_set}
    matches = 0
    for f in top:
        if f in ref or any(f.startswith(r + "_")
                           for r in ref):
            matches += 1
    return matches / k
```

Cross-Dataset Canonical Feature Alignment

Maps TON_IoT and CIC-IoT-2023 to 6 shared features for the cross-dataset experiment (Section 3.10.1, Table 3.1).

```
CANONICAL_FEATURES = [
    "f_total_pkts", "f_total_bytes", "f_duration",
    "f_is_tcp", "f_is_udp", "f_is_icmp",
]

def align_ton_iot(df):
    out = pd.DataFrame(index=df.index)
    out["f_total_pkts"] = df["src_pkts"] + df["dst_pkts"]
    out["f_total_bytes"] = df["src_bytes"] + df["dst_bytes"]
    out["f_duration"] = df["duration"]
    proto = df["proto"].str.lower()
    out["f_is_tcp"] = (proto == "tcp").astype(int)
    out["f_is_udp"] = (proto == "udp").astype(int)
    out["f_is_icmp"] = (proto == "icmp").astype(int)
    out["label"] = df["label"]
    return out
```

High-Cardinality Feature Filter

Prevents one-hot encoding of high-cardinality columns (e.g. `dns_query` with hundreds of unique hostnames) that would inflate the feature space (Section 3.3).

```
def _high_cardinality(series):
    n = len(series)
    if n == 0:
        return False
    nunique = series.nunique(dropna=True)
    return nunique > 50
```